ETH zürich | SPACE



PROJECT PHENEI

Last Mile Lunar Hopper

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Chapter 1

Executive Summary

[ilya: I think this part should be rewritten once]

More than half a century after the last Apollo landing, the Moon is once again a destination, and this time a commercial one. Lunar traffic is expected to grow rapidly: industry estimates anticipate up to 400 lunar missions over the next 20 years **[reference]**, and the lunar economy is projected to exceed €142 billion by 2040 **[reference]**.

Commercial lunar delivery is splitting into two markets. Small landers such as Blue Ghost and Nova-C can place payloads almost anywhere on the Moon, but cost more than €1 million per kilogram of payload **[reference]**. A second marked now in development are large and mega-class landers, who are expected to cut this cost to as little as €15,000 per kilogram. However, they will carry only large payloads on infrequent flights, mostly to the lunar South Pole, whereas many customers need to go elsewhere. Interlune, for example, is targeting the nearside equator, where helium is most abundant in the regolith, more than 2,700 km from the pole.

PHENEI is designing a Last Mile Hopper to close this gap. The hopper rides as rideshare cargo on a large lander to the Shackleton Crater connecting ridge. After offloading, it flies a single autonomous propulsive hop to its final destination, with operations supported by Interlune. Once the hopper has landed, the payload customer takes over operations.

The reference customer is Interlune's Prospect Moon. It aims to be the first to measure volatile gases by heating lunar regolith on the Moon itself. A robotic arm collects regolith, which is sorted and heated to release solar-wind-implanted gases such as helium-3 and hydrogen for analysis by a mass spectrometer. The results will also show how much power full-scale helium-3 extraction requires. Interlune prefers equatorial landing sites and does not consider the South Pole, where the large landers will deliver, a preferred location. To host the payload, the hopper provides 200 W of continuous power during the lunar day, surface access for the robotic arm, and at least 2 Mbps of data downlink to Earth.

PHENEI pursues three main objectives. The first is to prove that a last-mile delivery system can operate under the harsh environmental conditions on the lunar surface. The second is adaptability: a modular design allows the hopper to serve future customers with different payload masses, power and data needs, and destinations, not only Prospect Moon. The third, and most important, is to demonstrate that last-mile delivery is possible for well under $1 million per kilogram of payload. This would give private customers a cheaper alternative to a dedicated small lander, and turn low-cost deliveries to the South Pole into affordable access to sites far beyond it.

Chapter 2

Mission Objectives (Ilya)

2.1 Why a Last-Mile Hopper

Access to the lunar surface is expected to shift from small landers (200 kg payloads) to heavy-lift cargo vehicles like Starship (100T payloads) in the coming decade. Accounting for the plans of NASA and SpaceX, landing site of cargo ships will be the lunar South Pole, while a lot of smaller payloads need precise access to places all over the surface and don't have capability to go there (tiny rovers have range of 15km around the landing site). While this leap in capacity will enable a lower cost of access, massive landers are ill-suited for delivering smaller payloads to more precise and potentially challenging final destinations. This is a similar challenge to satellite rideshare missions to LEO, which leave individual payloads to rely on other propulsion methods to reach their final desired orbit.

To solve this bottleneck, we design a modular lunar hopper architecture capable of safely transporting payloads from a primary heavy lander's touchdown site to their final destinations. Rather than a single-task vehicle,our system targets versatile use providing service to different commercial customers, and agnostic to the potential landing sites.

To demonstrate this modular architecture, the project is focused on a primary mission to deploy a suite of site survey and resource prospecting instruments on the equator of the moon. Delivering this payload across diverse sites will prove the hopper's mobility and flexibility while acting as a scout for future In-Situ Resource Utilization (ISRU). While mapping local resources and topography, this mission will demonstrate how a last mile delivery platform enables leveraging large, economical landers to access the lunar surface, without sacrificing the precision of the final destination.

2.2 Mission and Objectives

The purpose of PHENEI is to deliver payloads from the landing site of a heavy cargo lander at the lunar South Pole to their final destination on the lunar surface, and to support their operation there. The reference mission is the delivery of the Interlune Prospect Moon payload to the lunar equator. The requirements derived from this mission are listed in Table 2; the objectives below summarise them.

2.2.1 Primary Mission Objectives

The primary objective is to deliver the Interlune Prospect Moon payload safely from the south polar landing site to the lunar equator and to support its operation at the destination.

Transport. The hopper departs from the rim of the Shackleton Crater connecting ridge (approx. 89.53° S, 150.05° W) and delivers at least 45 kg of payload to any location on the lunar equator (M1.1). All flight phases from liftoff to touchdown are executed autonomously, without real-time ground intervention (S1.1.1), and the landing is completed within 100 m of the target site (S1.1.3).

Payload support. At the destination the hopper serves as the platform for the payload. It mounts the instruments and sampling devices, such as a small robotic arm, so that they view and reach the lunar surface (M2.2). It provides at least 200 W average power during lunar day (M2.1), transmits at least 2 Mbps of payload data to Earth whenever ground stations or relay satellites are in line of sight (M2.3), and supports payload operations for at least 72 h (M2.4).

Modularity and cost. The hopper should accommodate at least two distinct payload configurations (M1.2), so that the same platform can serve customers other than Interlune. The estimated cost per kilogram of delivered payload shall remain below 1 M$/kg (M4.1).

2.2.2 Additional Mission Elements

In addition to the primary payload, the hopper should offer transport to secondary customer payloads (P3). Rideshare is offered exclusively at selected drop-off points along the primary trajectory. Secondary payloads of up to 10 kg in total (M3.1) are released at their assigned site on command (M3.2, S3.2.1). The drop-off sites, the number of secondary payload positions and their accommodation are not yet defined; the analysis and a proposed solution will follow after 14 November 2026.

2.3 Stakeholder Analysis

Table 1: Stakeholder analysis for PHENĒI.

Stakeholder	Role	Interests/Objectives	Benefits
Customers			
Interlune	Primary customer	$\geq$ 50 kg of science payload, delivered safely to a specific site. $<$ €1M/kg, beating a dedicated small lander. Open to any customer within range.	Prospects He-3 at the lunar equator. Reaches site-specific destinations without buying a dedicated lander.
ispace, ETH Zürich, NASA	Secondary customer	Delivers probes of $\leq$ 10 kg. Drop-off point within ~100–300 km of the primary customer's trajectory start or end. Soft drop-off or release in flight.	Reach sites beyond own range. Rideshare price. Power from the hopper.
Operator			
PHENĒI	Operator	Deploys from Starship. Delivers primary and secondary payloads to their sites. Provides power, data and a thermal path until end of mission.	Lowest €/kg. Sells spare capacity to secondaries. Demonstrates a reusable "last-mile" business.
Launch Provider			
SpaceX	Launch provider	Delivers ~100 t to the lunar south pole. Offloads Hopper via the cargo elevator.	Fills Starship capacity. $100k/kg now. Expected to drop towards ~€15k/kg.
End Users			
Interlune	End user	He-3 resource data from the equator. Seismic and rover data from secondary payloads.	Validates the resource case for ISRU. Science return per €.
ESA, NASA, Lunar Leaper	Future end user	Routine point-to-point surface logistics.	Precise site access. No dedicated lander needed.

Chapter 3

Mission Requirements (Ilya)

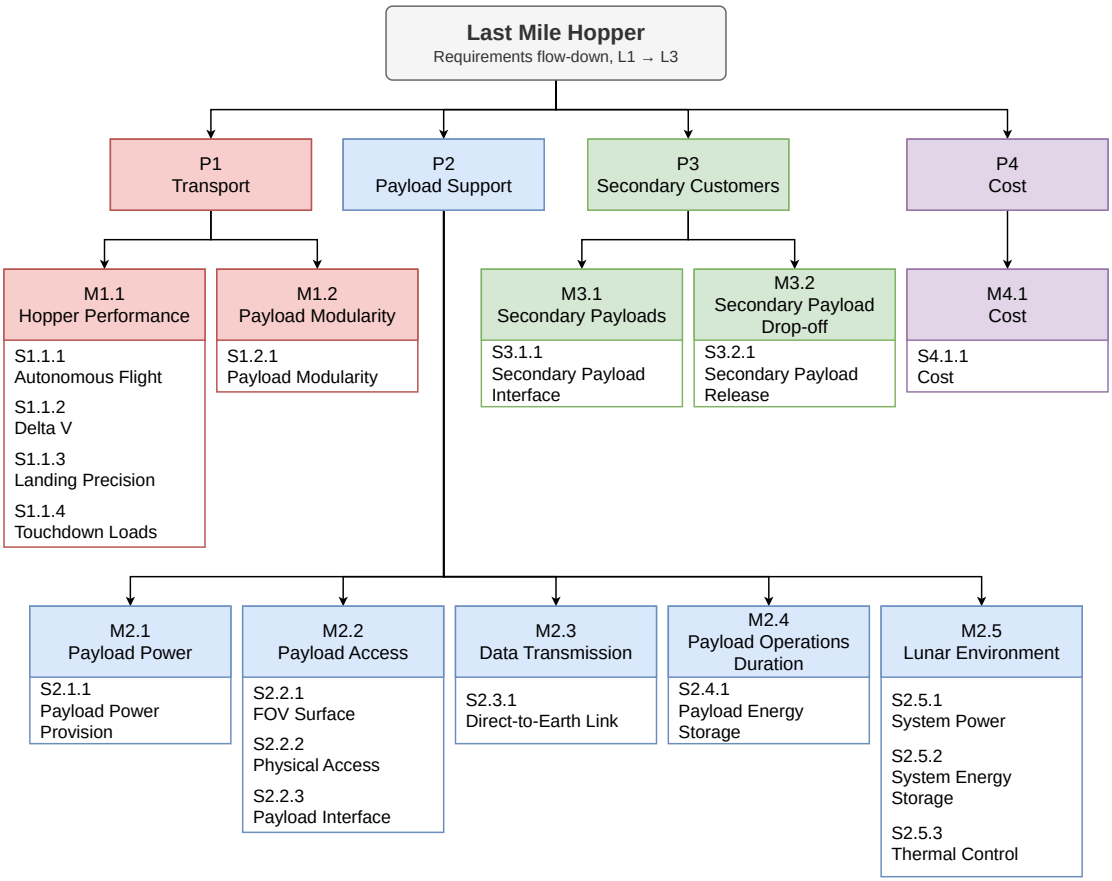


Table 2: Requirements of the Last Mile Hopper, levels L1 (programmatic, P), L2 (mission, M) and L3 (system, S). The ID gives the level letter followed by the numbers of all parent requirements and an index, so S2.5.1 derives from M2.5, which derives from P2. Type: *Shall* is mandatory, *Should* is a goal. K/D: K marks a key requirement, needed to meet the mission objectives; D marks a driving requirement, one that is particularly difficult, costly or challenging to meet. Value: the quantity that verifies the requirement; TBD and TBC mark values still to be determined or confirmed. Rationale: why the requirement exists, where it is not self-evident.

ID	Type	K/D	Title	Requirement	Value	Rationale
L1 — Programmatic						
P1	Shall		Transport	The Last Mile Hopper shall transport the payload within a nominal volume and mass of Prospect Moon Mission from the south-pole landing site to its destination in the equatorial region.		

ID	Type	K/D	Title	Requirement	Value	Rationale
P2	Shall		Payload Support	The Last Mile Hopper shall support the operation of the main payload at its destination.		
P3	Should		Secondary Customers	The Last Mile Hopper should offer the transport to secondary customer payloads along the primary trajectory.		
P4	Shall		Cost	Estimated cost per kilogram of payload shall be below the cost of currently available lunar landers	1M$/kg	Programatic requirement, the The Last Mile Hopper is not feasible unless fulfilled. Estimation involves simple formulas provided by primary shareholder and making the MR the driving requirement
L2 — Mission						
M1.1	Shall	K, D	Hopper Performance	The Last Mile Hopper shall be capable of delivering at least 45 kg of payload from the rim of Shackleton Crater Connecting Ridge (approx. 89.53° S, 150.05° W) to any location on the lunar equator.	≥ 45 kg	
M1.2	Should		Payload Modularity	The Last Mile Hopper should be capable of accommodating at least 2 distinct payload configurations as defined in the Payload Configuration Table.	≥ 2 configurations	
M2.1	Shall	K	Payload Power	The Last Mile Hopper shall be capable of supporting payloads with at least 200 W average power during lunar day.	≥ 200 W	
M2.2	Shall	K	Payload Access	The Last Mile Hopper shall be capable of mounting and accommodating payload instruments and sampling devices that view and reach the lunar surface.		
M2.3	Shall	K	Data Transmission	The Last Mile Hopper shall be capable of transmitting at least 2 Mbps of data from the customer payload back to Earth whenever within line of sight of Earth ground stations or lunar relay satellites.	10 > Downlink ≥ 2 Mbps	
M2.4	Shall	K	Payload Operations Duration	The Last Mile Hopper shall support payload operations at the destination for at least 72 h.	≥ 72 h	
M2.5	Shall		Lunar Environment	The Last Mile Hopper shall operate within specification in the lunar surface environment from Shackleton Crater to the destination for at least TBD h.	≥ TBD (roughly 1hr) h	While doing a hop we have variable power from solar panels, our avionics and navigation system, particularly image based map matching causes power spike which shall be accounted.
M3.1	Should		Secondary Payloads	The Last Mile Hopper should carry secondary payloads of up to 10 kg in total along the primary trajectory	10kg	
M3.2	Should		Secondary Payload Drop-off	The Last Mile Hopper should drop off each secondary payload at its TBD site along the primary trajectory.	TBD possible sites	
M4.1	Shall	K	Cost	Estimated cost per kilogram of payload shall be below 1M$/kg	1M$/kg	

ID	Type	K/D	Title	Requirement	Value	Rationale
L3 — System						
S1.1.1	Shall	D	Autonomous Flight	The Last Mile Hopper shall execute all flight phases from liftoff to touchdown without real-time ground intervention.		Autonomy is necessary for the hop due to the absence of reliable communication networks in lunar environment. E.g. we cant rely on any land-based antena while landing in the crater
S1.1.2	Shall	D	Delta V	The Last Mile Hopper shall provide a total ΔV capability of at least 3 km/s (TBC) while carrying a 45 kg payload.	≥ 3 km/s (TBC)	
S1.1.3	Shall	D	Landing Precision	The Last Mile Hopper shall complete its landing procedure within a 100 m radius of the target landing site.	≤ 100 m	
S1.1.4	Shall		Touchdown Loads	The Last Mile Hopper shall withstand quasi-static accelerations of at least TBD m/s² axial and TBD m/s² lateral without structural failure or loss of function.	TBD / TBD	
S1.2.1	Should		Payload Modularity	The Last Mile Hopper should provide a payload mounting surface with several mounting points over a usable area of at least TBD m²	TBD m²,	Allowes payload configurations to be changed without hardware modification of the hopper.
S2.1.1	Shall		Payload Power Provision	The Last Mile Hopper shall provide the payload with 200 W average and TBD W peak electrical power at the destination.	200 W avg / ≥ 300 W peak (TBC)	
S2.2.1	Shall		FOV Surface	The Last Mile Hopper shall provide hosted payloads with an unobstructed field of view of the lunar surface of at least TBD.	TBD	
S2.2.2	Shall		Physical Access	The Last Mile Hopper shall provide hosted payloads with physical access to the lunar surface within a TBD m radius of the payload position.	TBD m	
S2.2.3	Shall		Payload Interface	The Last Mile Hopper shall provide an identical payload interface at every payload accommodation position, as specified in the Payload ICD.	28 V payload bus	
S2.3.1	Shall		Direct-to-Earth Link	The Last Mile Hopper shall be capable of transmitting data from the Moon's surface to Earth ground stations through a direct connection.	10 > Downlink ≥ 2 Mbps	
S2.4.1	Shall		Payload Energy Storage	The Last Mile Hopper shall provide at least TBD Wh of electrical energy to hosted payloads over the 72 h of payload operations at the destination	≥ TBD Wh , 72 h	
S2.5.1	Shall		System Power	The Last Mile Hopper shall provide its own subsystems with 200 W average and TBD W peak electrical power.	200 W / TBD W	
S2.5.2	Shall		System Energy Storage	The Last Mile Hopper shall be capable of storing enough energy to sustain TBD min of continuous operation.	≥ TBD min	
S2.5.3	Shall	D	Thermal Control	The Last Mile Hopper shall keep all its thermal zones within their operating temperature limits during all its phases starting from launch and ending at end of life.	TBD to TBD °C per thermal zone	Conditions range from low sun at the pole to equatorial noon, plus engine heating during the hop.

ID	Type	K/D	Title	Requirement	Value	Rationale
S3.1.1	Should		Secondary Payload Interface	The Last Mile Hopper should provide at least TBD secondary payload positions with the interface specified in the Payload ICD.		
S3.2.1	Should		Secondary Payload Release	The Last Mile Hopper should release a secondary payload of up to 10 kg onto the lunar surface on command.	≤ 10 kg	
S4.1.1	Shall	D	Cost	Mass Ratio of the The Last Mile Hopper should be more than TBD %	TBD %	

TODO: K/D table

Chapter 4

Mission Design

4.1 Mission Design Drivers (Freddy)

The figure of merit of the mission is the cost per kg of payload delivered to the target site. The following drivers dominate the mission design through their effect on this figure of merit.

[Host lander constraints (start point, arrival time, price per kg, interfaces) are distributed among the drivers below and collected in Section Launch and Delivery System, instead of forming a separate driver. Feedback welcome on whether this allocation is preferable to a dedicated driver.]

Distance and Δ V: Distance → required Δ V → exponential growth of propellant mass (rocket eq.) → larger tanks → higher dry mass → more propellant (mass snowball) → higher mass delivered by host lander → higher cost per kg payload.

Source: **[L1]** *Affects:* **[TBD]**

Lunar Day Timing: Lunar day-night cycle → large swings in solar irradiance and surface temperature → night survival requires batteries, heaters and insulation → higher dry mass → more propellant and higher delivery cost → higher cost per kg payload.

Without night survival: arrival time set by host lander → transfer, landing and payload operations must fit into the remaining daylight → limited operational time window.

Source: Environment, host lander constraint *Affects:* **[TBD]**

Landing on Unprepared Terrain: **[unsure if this is actually a design driver]**

Source: **[TBD]** *Affects:* **[TBD]**

Earth Visibility: Earth below local horizon or blocked by terrain → no direct-to-Earth link → relay (host lander or orbiter), onboard data storage or waiting for visibility → added mass, relay service cost or constrained operations.

Ground commanding only within visibility windows → higher onboard autonomy or restricted operational timeline.

Source: Environment *Affects:* **[TBD]**

Payload Diversity: Customers differ in payload mass, volume, power and data needs → hopper sized for the upper end of the envelope → unused capacity on most missions → higher cost per kg for smaller payloads.

Payload interfaces must accommodate the full range of needs → power, data and mechanical provisions sized for the most demanding payload → higher dry mass → higher cost per kg payload.

Source: **[L1]** *Affects:* **[TBD]**

4.2 Mission Trade-offs (Freddy)

4.2.1 Alternative Concepts

Mission Timeline

The lander operator sets the delivery date, and the lander is expected to touch down on the Shackleton crater rim at the start of the longest illumination period. Whether the hopper's destination, located closer to the equator, is lit at that time depends on its longitude relative to the subsolar point, so it may be in lunar night on arrival. This leads to two alternative mission timelines. In the first, the hopper waits in a low-power standby mode near the offload site, possibly after a short repositioning hop out of the lander's shadow and the offloading zone (see trajectory trade-off). It recharges its batteries in sunlight and performs the transfer hop once the destination is illuminated. This allows a daylight landing with optical navigation, but requires long-term propellant storage. That favours storable propellants and an isolated pressurant supply over cryogenic ones. In the second, the hopper flies to the destination directly after deployment and enters hibernation until the night is over. This minimises storage time but requires a landing in darkness and survival of the night without solar power, including keeping the propellant above its freezing point, which drives battery, heater and thermal mass. A hybrid option, with an intermediate stop at an illuminated site, is assessed in the hop trade-off. The waiting strategy is currently preferred as the baseline, as it avoids the two most critical risks of a night landing and night survival.

Communication

There are mainly to options for the communication concept, since a relay via the cargo lander is not assumed, as it cannot be guaranteed as a payload service. Two options remain. The first is a direct-to-Earth (DTE) link, which is simple and independent of third parties but requires Earth visibility and higher transmit power. The second is a lunar relay satellite, which needs only low power and also reaches the far side and terrain-shadowed polar sites, but offers limited contact windows and depends on a service provider. Relay systems such as ESA's Moonlight and NASA's Near Space Network are planned for the 2030s, but no specific provider can be assumed today. The choice is therefore driven by the expected customer destinations. For Prospect Moon Interlune prefers equatorial landing sites, and helium-3 is associated with titanium-rich mare regolith, found mainly on the near side. **[finde source for that]** DTE is therefore selected as the primary link, as it fully covers such near-side, equatorial destinations. The hopper is additionally designed to be relay-compatible, to improve coverage at the south pole and to enable far-side destinations should a customer require them. Since the hop is executed autonomously, no real-time link is required during the hop: data is stored onboard and downlinked at the next contact.

Autonomous vs Man Controlled

[needs to be decided if its worth discussing]

Trajectory

[needs to be defined what to put here and what to put into the rajectory design chapter in 7]

4.2.2 Alternative Architecture

Transport Mode

An alternative architecture would be to use a rover instead of a hopper. Without a propulsion system there would be no launch and landing phases, which removes a major risk. The system mass would also drop drastically, and the savings could be used for heavier or additional payloads. The weight savings could be **[define the weight savings]**. However, a rover has major disadvantages. It is bound to the surface, so every mountain and valley has to be climbed or descended instead of flown over. This carries a significant risk of tipping over or getting stuck, likely higher than the risk of a hopper's launch and

landing, since lunar terrain cannot be fully replicated or tested on Earth. A rover is also very slow. At around 1 km/h [1] , reaching the equator about 2,700 km away would take more than four years of pure driving, ignoring stops and lunar nights. Few customers would accept such delivery times, especially as lunar traffic and development are expected to grow rapidly. Operating a rover for several years would most likely cost more than the mass savings bring in **[some reasoning/calculation]**.

4.3 Product Tree (Ilya)

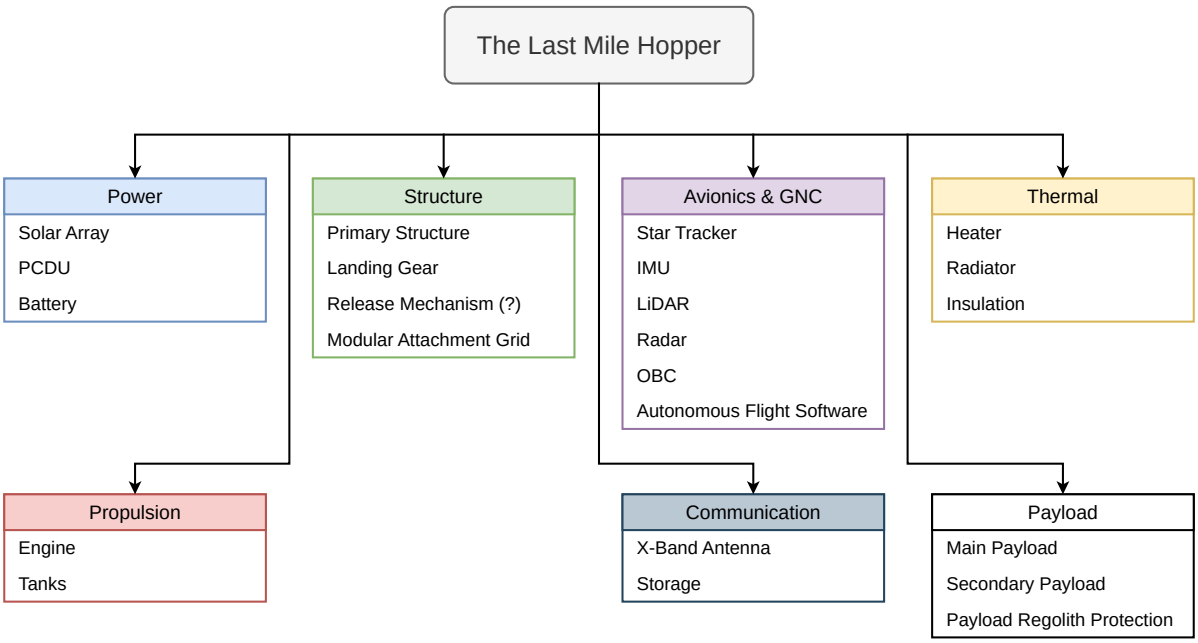


Figure 1: …

The release mechanism is currently marked as (?), it as decision about secondary payloads will be made after 14.11.

4.4 Interface Descriptions (PDR Ilya)

[Hopper–carrier and hopper–payload interfaces.]

4.5 Budgets (Ilya)

Table 3: System budget of the Last Mile Hopper. Masses of the navigation sensors and the on-board computer are taken from Table 7 (incl. 5–20 % margin); all other subsystem masses are a first allocation from the planetary mass fractions in [2, Table 14-18], scaled from the 45 kg payload, without margin. Power: highest phase average (continuous) and highest worst-case peak over all mission phases from Table 10. A dash (–) means the subsystem is not listed there.

Subsystem	Mass [kg]	Peak power [W]	Peak power duration [min]	Peak heat [W]	Memory	Continuous power [W]	Source / owner
Payloads	**45**						
Prospect Moon payload (Interlune)	45	250	TBD	TBD	TBD	200	M1.1, M2.1; P-01, P-05
Secondary payload (≤ 10 kg)	in payload	TBD	TBD	TBD	TBD	TBD	M3.1; P-04

Subsystem	Mass [kg]	Peak power [W]	Peak power duration [min]	Peak heat [W]	Memory	Continuous power [W]	Source / owner
Propulsion & Attitude Control	**39**	**400**				**200**	
Main engine	TBD	TBD	TBD	TBD	TBD	TBD	Mattia (WBS 4)
RCS / TVC	TBD	TBD	TBD	TBD	TBD	TBD	Mattia (WBS 4.4)
GNC, Avionics & Communications	**25**						
Navigation sensors	2.13	180	TBD	TBD	TBD	150	Nicola (WBS 5.1)
On-board computer & data handling	1.43	50	TBD	TBD	TBD	40	Nicola (WBS 5.3)
Communication	21	100	TBD	TBD	TBD	100	Nicola (WBS 5.4)
Power & Thermal	**81**						
Power (generation, storage, distribution, harness)	63	20	TBD	TBD	TBD	20	Fabian (WBS 6.1)
Thermal control	18	400	TBD	TBD	TBD	200	Fabian (WBS 6.2)
Structure & Mechanisms	**75**						
Structure & mechanisms	75	–	–	–	–	–	Lorenzo (WBS 7)
Other	**9**						
Balance + launch	9						
Dry mass (no system margin)	**274**	**970**				**630**	Power: landing phase, Table 10
Propellant	**821**						ΔV 4.35 km/s (P-09), I_{sp} 320 s (P-11)
Wet mass	**1094**						

4.6 Technology Readiness Assessment (PDR Fabian)

[TBD]

4.7 Compliance Matrix (PDR Ilya)

[Each requirement from vs. design status.]

4.8 AIT Plan (PDR Fabian)

[TBD]

4.9 Risk Assessment (Fabian)

Potential Risks:

- Program:
  - Lunar Starship delay
  - Impossibl to reach low price
  - no customers
- Propulsion:
  - Leakge of Pressurization
  - No Ignition
  - Freezing Valves/Sensors

- AV/GNC:
  - Navigational errors (Crash)
  - poor lighting -> hard to navigate
- Power:
  - operation extensions (lunar night)
- Structure:
  - rough environment (stability)
  - hard landing
  - offloading from starship

Chapter 5

Spacecraft Design

5.1 System Trade-offs

5.2 Propulsion Subsystem (Mattia)

[includes: piping, valves, engine, cad, where is what located, tank design and placement, propulsion related pressure and temperature sensors, maybe pumps; gets the thrust curve from mission design and gives mission design the massflow]

[General intro, drivers, trade-offs, design]

Propulsion choice is a key factor in The propulsion subsystem is one of the most important elements of the overall system design. It is responsible for ensuring that the Hopper, starting from the Lunar South Pole, can reach its target destination on the Lunar Equator as cheaply and reliably as possible. This means accomplishing both ballistic trajectory insertion through high-thrust ΔV maneuvers and attitude control low-impulse corrections using high-authority, finely throttled thrusters. Since engine-specific characteristics such as thrust curves and throttleability constrain trajectory optimization, tank and thruster envelopes affect structure and mass distribution, and operating temperature ranges drive thermal requirements, the propulsion subsystem has to be designed with the whole system in mind - building the engines around the mission, not vice versa. Finally, given that propellant and propulsion hardware constitute the vast majority of total spacecraft mass, implementing full system redundancy is unfeasible; reliability must therefore be achieved intrinsically by reducing complexity and leveraging flight-proven technology to the extent possible. Nevertheless, localized redundancy is implemented at the piping, regulation, and valve levels to minimize Single Points of Failure (SPOFs).

5.2.1 Key Drivers

Among the main drivers to enumerate, I_{sp} is the first - and arguably the single most important - performance metric of a propulsion system. However, I_{sp} alone does not tell the whole story: throttleability, attitude control, restart capability, simplicity, total mass and volume envelopes, and overall cost all play a crucial role in determining the best-suited architecture for the mission.

For this reason, it is of paramount importance to establish a concrete framework to evaluate competing propulsion architectures and rationalise the final selection. To tackle this, a weighted evaluation system is introduced: to each key driver is assigned an importance weight (0–10) relative to overall mission requirements. Candidate architectures are then scrutinised and assigned a score (0–10) for each driver relative to competing options. The architecture that accumulates the highest total weighted score is selected. Note that these scores hold no absolute physical meaning and are only valid within the context of this comparative trade. Consequently, including architectures that are obviously ill-suited would artificially inflate scores and skew the results; sound engineering discretion must be exercised.

ID	Driver / Parameter	Weight	Rationale
Performance			
KD-01	Throttleability	8	Enables soft touchdown and hover precision while limiting impact loads on landing gear
KD-02	Vectoring and attitude control	8	Provides active orientation control and thrust vector alignment during manoeuvres
KD-03	Restart and multi-hop capability	7	Enables sequential engine ignitions across multi-burn ballistic trajectories and hopping ops
KD-04	$T_{\max}$	6	Enables high thrust-to-weight ratio to minimise gravity losses during ballistic burns
	Performance Subtotal:	**29**	
Mass Constraints			
KD-05	I_{sp}	10	Drives required propellant mass, constituting the majority of initial wet mass
KD-06	Subsystem dry mass	10	Constrains dry hardware mass (tanks, feed system, engines) independently of I_{sp}
	Mass Constraints Subtotal:	**20**	
Reliability			
KD-07	Simplicity	7	Minimises component count, single points of failure (SPOFs), and integration risk
KD-08	TRL	6	Prioritises flight-proven space hardware to mitigate development schedule risk
	Reliability Subtotal:	**13**	
Other Systems Constraints			
KD-09	Thermal constraints	8	Limits operational temperature bounds to prevent propellant freezing or boil-off
KD-10	Total volume	5	Constrains physical envelope to fit within launcher payload bay and structural deck bounds
	Other Systems Constraints Subtotal:	**13**	
Business Constraints			
KD-11	Cost	5	Limits non-recurring development and recurring unit procurement expenditure
KD-12	ROI	3	Evaluates return on investment and commercial viability for future mission extensions
	Business Constraints Subtotal:	**8**	
	Total Weight	**83**	

Table 4: Key driver weights and trade-off criteria.

5.2.2 Architectures: Candidate Options, Preliminary Exclusions, and Selection

The main candidate propulsion architectures evaluated for the system are:

- **MON-3 + MMH:** A hypergolic liquid bipropellant architecture utilising Mixed Oxides of Nitrogen (MON-3, containing 3% Nitric Oxide by mass in Nitrogen Tetroxide) as the oxidiser and Monomethyl-

hydrazine (MMH) as the fuel. It possesses extensive flight heritage (including the Apollo Lunar Module engines and modern lunar landers like Firefly Blue Ghost **[Add Orion/FBG sources]**), seamless dual-mode Reaction Control System (RCS) integration, high hypergolic restart reliability, high $I_{\text{sp}}(\approx 315 - 330\ \text{s})$ and a relatively low freezing point ($-15\ ^\circ\text{C}$) for surviving lunar thermal dips **[Add architecture sources]**.

- **HTP + RP-1:** A non-toxic, catalytic liquid bipropellant architecture employing High-Test Peroxide (HTP, $> 98\%$ concentration **[Add sources]**) as the oxidiser and Refined Petroleum-1 (RP-1) kerosene as the fuel. Passing HTP over a precious-metal catalyst bed decomposes it into superheated steam and oxygen, auto-igniting the injected RP-1. This yields high specific impulse ($I_{\text{sp}} \approx 305 - 315\ \text{s}$) and high bulk propellant density, while allowing oxidiser tap-off for monopropellant RCS thrusters **[Add architecture sources]**.
- **HTP + Paraffin / HDPE:** A green hybrid propulsion architecture pairing liquid HTP ($> 98\%$ concentration **[Add sources]**) as the oxidiser with a solid hydrocarbon fuel grain (such as High-Density Polyethylene or liquefying Paraffin wax). The catalytically decomposed hot HTP auto-ignites the solid grain, enabling single-valve fluid control, passive reignition without electrical igniters, and reduced subsystem complexity by eliminating liquid fuel tanks and plumbing **[Add architecture sources]**.
- **MON-3 + HTPB:** A storable hybrid architecture combining a high-density liquid MON-3 oxidiser with a solid Hydroxyl-Terminated Polybutadiene (HTPB) synthetic rubber grain. While benefiting from storable thermal bounds and low tank complexity, non-hypergolic reignition requires active secondary ignition systems for every restart, and deep throttling suffers from severe Oxidiser-to-Fuel (O/F) ratio shifts **[Add architecture sources]**.

Other propulsion architectures commonly employed in space missions were preliminary excluded from detailed trade evaluation due to fundamental operational limits:

- **Solid Propellant Systems:** Solid rocket motors offer mechanical simplicity, volumetric loading efficiency, high thrust-to-weight ratios ($\text{T/W} > 100$), and long storable shelf-life — leveraging proven technologies such as Ammonium Perchlorate Composite Propellant (APCP), star-grain core geometries, and flex-bearing Thrust Vector Control (TVC) nozzles. However, they were excluded due to the physical impossibility of throttling thrust or shutting down and reigniting the grain, which are non-negotiable requirements for soft lunar landings and multi-hop trajectories **[Add architecture sources]**.
- **Cryogenic Propellant Systems:** Cryogenic combinations - primarily Liquid Oxygen (LOX) paired with Liquid Hydrogen (LH_2) or Liquid Methane (LCH_4, methalox) - deliver very high chemical efficiency ($I_{\text{sp}} > 380 - 450\ \text{s}$) and clean, non-toxic combustion products. Nevertheless, maintaining cryogenic storage temperatures ($\sim 90\ \text{K}$ for LOX, $\sim 20\ \text{K}$ for LH_2) over extended surface standby leads to severe boil-off losses and complex active cryocooling power demands that cannot be accommodated within a compact hopper mass budget **[Add architecture sources]**.
- **Electric Propulsion Systems:** Electrostatic and electromagnetic architectures — such as Hall Effect Thrusters (HETs), Gridded Ion Engines (GIEs), Pulsed Plasma Thrusters (PPTs), and electrospray arrays - achieve unmatched specific impulse ($I_{\text{sp}} > 1500 - 3000\ \text{s}$) and minimal propellant mass requirements. However, their extremely low thrust output (ranging from millinewtons to single-digit newtons) yields vehicle thrust-to-weight ratios far below unity ($\text{T/W} \ll 1$). Consequently, they are physically incapable of executing ballistic hop insertions, rapid trajectory adjustments, or soft touchdowns on the lunar surface **[Add architecture sources]**.

ID	Weight	MON-3+ MMH	HTP + RP-1	HTP+ HDPE	MON-3+ HTPB	Score Justification & Comments
Performance						
KD-01	8	9	8	5	4	Liquid bipropellants allow fine pintle throttling; hybrids suffer from O/F shift during throttling
KD-02	8	9	9	9	3	HTP acts as a monopropellant for RCS; MON-3 bi propellant integrates well with RCS; MON-3 hybrids require a completely separate RCS system
KD-03	7	10	8	7	2	Hypergolic MON-3 offers infinite restarts; MON-3 hybrid requires highly complex active ignition
KD-04	6	8	9	8	4	Liquids and liquefying paraffin scale easily; HTPB's slow regression limits thrust output
Subtotal:	**29**	***262***	***246***	***209***	***94***	
Mass Constraints						
KD-05	10	9	8	7	6	Bipropellants offer more performance than hybrids
KD-06	10	4	TBD	TBD	TBD	Bipropellants are heavy due to dual-tank plumbing
Subtotal:	**20**	***130***	***80 (TBC)***	***70 (TBC)***	***60 (TBC)***	
Reliability						
KD-07	7	3	3	8	7	Hybrids eliminate a liquid feed circuit, reducing valve and plumbing
KD-08	6	10	7	5	4	MON-3/MMH is the deep-space standard; hybrids possess minimal to zero lunar flight heritage
Subtotal:	**13**	***81***	***63***	***86***	***73***	
Other Systems Constraints						
KD-09	8	7	5	5	7	HTP freezes at −11°C and undergoes spontaneous decomposition, requiring more thermal management; MON-3 at −15°C
KD-10	5	7	9	6	4	HTP/RP-1 has highest bulk density; multi-port HTPB grains have poor volumetric filling
Subtotal:	**13**	***91***	***85***	***70***	***76***	
Business Constraints						
KD-11	5	TBD	TBD	TBD	TBD	
KD-12	3	TBD	TBD	TBD	TBD	
Subtotal:	**8**	***0 (TBC)***	***0 (TBC)***	***0 (TBC)***	***0 (TBC)***	
Total Weighted Score		**564 (TBC)**	**474 (TBC)**	**435 (TBC)**	**303 (TBC)**	

Table 5: Candidate architecture trade matrix and score evaluations. Each score derived from calculations and or modelling is TBC

As shown in Table 5, the MON-3 + MMH baseline holds a decisive lead over HTP + RP-1, despite several preliminary (TBC) and unassigned (TBD) scores. While ongoing trade study refinements and detailed performance modelling are expected to narrow this margin — particularly as metrics such as propulsion subsystem dry mass (KD-06) and commercial ROI (KD-12) are fully characterised for the HTP and hybrid alternatives, the current score differential is sufficiently substantial to confirm MON-3 + MMH as the baseline architecture for preliminary subsystem design. This selection is further reinforced by the fact that the primary runner-up (HTP + RP-1) is likewise a liquid bipropellant system; consequently, overall fluid feed routing, tankage layout, and structural integration requirements would remain broadly similar should a design pivot be required in later project phases. Consequently, it remains the primary candidate for detailed thermal, structural, and feed-system integration.

For an in-depth derivation of the volume and mass relationships used to characterise the chosen system refer to Appendix B

5.2.3 Propulsion System Design

To simultaneously accommodate the high thrust required for ballistic trajectory injection, the deep throttleability necessary for soft landings, and precise in-flight attitude control — all while minimising system mass and complexity — the following propulsion architecture is proposed:

- TBD Nammo LEROS 4-class main engines (nominally TBD N each) will be utilised primarily to execute the high-impulse main ballistic burns.
- TBD Reaction Control System (RCS) thrusters (nominally TBD N each) will be employed to assist the descent burn and provide the fine, high-authority throttleability required for terminal touchdown.
- TBD smaller cold-gas Attitude Control System (ACS) thrusters will be distributed across the vehicle structure to ensure full in-flight orientation control and precise thrust vector alignment.

Both the main engines and the RCS thrusters will draw from a shared MON-3 and MMH propellant feed system, reducing plumbing complexity and redundant tankage mass. Furthermore, to eliminate the dry mass penalty of an independent cold-gas storage architecture, the ACS thrusters will directly tap into the main propellant pressurant supply (Helium, TBC). This highly integrated approach provides reliable, low-impulse attitude control while minimising discrete component count and potential single points of failure.

[Add subsystem-specific L4 requirements, and add a schematic of the propulsion system comprising plumbing and valves]

5.3 Avionics & GNC Subsystem

5.3.1 Overview and design drivers

The Avionics & GNC subsystem acquires data on the state of the Hopper and its environment, processes it on board and commands the other subsystems (propulsion, power, thermal control and payload) according to the mission phase. It also stores the payload data and hands them to the Communications subsystem.

This subsystem, and above all its testing, is critical to mission success. Several recent commercial landing failures were not caused by faulty sensors: the navigation chains had not been verified end-to-end under representative conditions (lighting, regolith albedo, lidar back-scatter). On IM-1 the laser rangefinders were unusable because a safety switch had not been enabled before launch [3]. On IM-2, altimeter interference and the low-angle South Pole lighting led to a landing on the side [4]. The subsystem also drives several ConOps decisions, such as illumination windows and the level of autonomy. The architecture is therefore built on two principles:

- **redundant COTS units with majority voting**, for the on-board computers and the IMUs;

- **dissimilar redundancy**: the same quantity is measured by sensors based on different physical principles, e.g. altitude from laser altimeters and a radar altimeter.

The main design drivers are:

- **Autonomy.** The hop is flown fully autonomously on a preloaded trajectory; during the hop the ground can only command abort or safe mode (AVI-23). Earth visibility is not guaranteed, and no lunar navigation satellite system is assumed at the equator (TR-01). Navigation therefore relies on the IMUs, the star trackers and terrain-relative navigation (TRN).
- **Landing precision.** Touchdown must be within a circle of 100 m diameter around the target (GNC-01), the precision of current commercial landers using camera-based TRN.
- **Safe touchdown.** Altitude and ground-relative velocity must be known down to touchdown, with at least two dissimilar altitude sensors (GNC-06, GNC-10).
- **Low cost.** Equipment is flight-proven or TRL $\geq$ 5 (AVI-11). COTS units are accepted where redundancy masks their weaknesses (AVI-OBC-02).
- **Short mission.** The subsystem must operate for 72 h at the destination during lunar day; night survival is not required (AVI-07).
- **Payload data.** Payload data are stored on board ($\geq$ 100 GB, AVI-05) and forwarded to Communications for the 2 Mbps downlink.

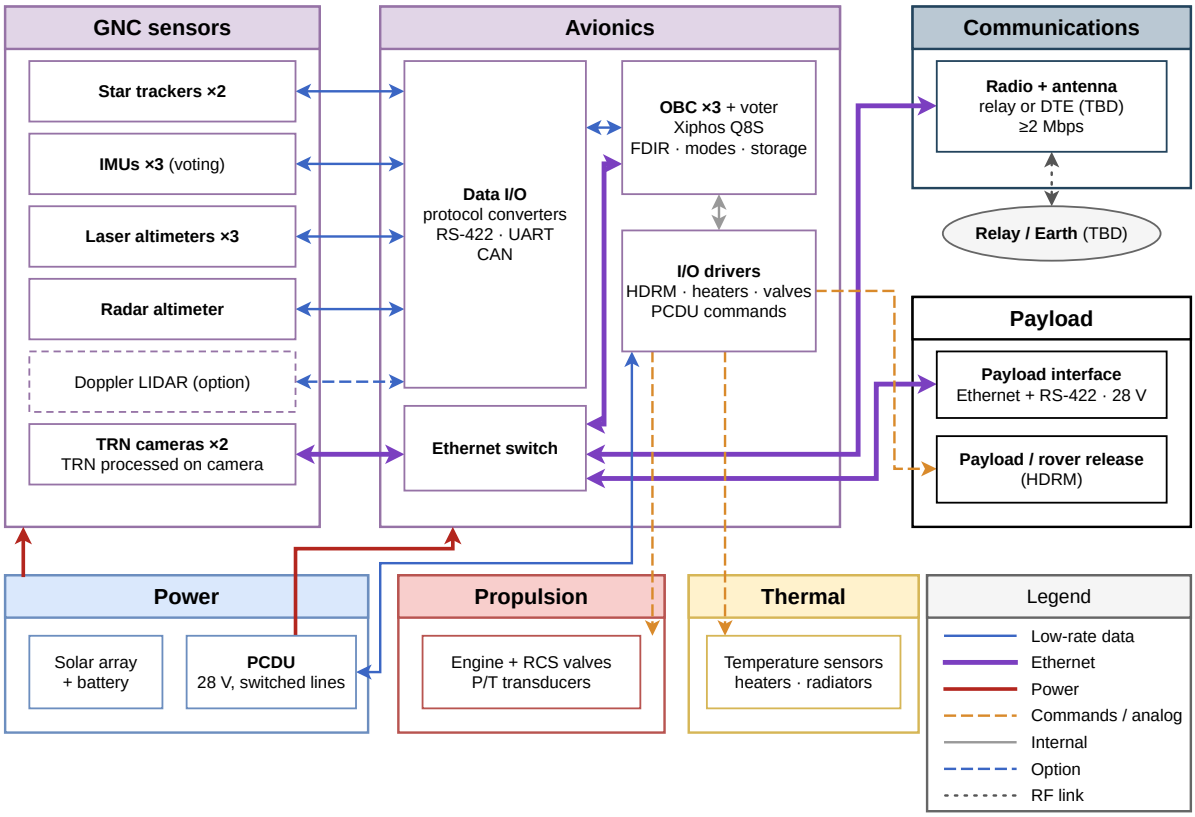


Figure 2: Avionics and GNC architecture (simplified).

Figure 2 shows the resulting architecture:

- **Low-rate sensors.** The GNC sensors are low-rate (IMU $\leq$ 2 kHz, laser altimeters $\leq$ 25 Hz, radar altimeter $\approx$ 60 Hz, star trackers 4 Hz). They are connected point-to-point (RS-422, UART, CAN) through protocol converters on the I/O board.
- **On-board computers.** Three Xiphos Q8S boards run with majority voting.

- **High-rate network.** Camera, payload and radio traffic uses an Ethernet switch, the natural choice since the Q8S offers Gigabit Ethernet but no SpaceWire (TR-11).
- **Vision processing.** TRN image processing runs on the processors of the Argus cameras (TR-12), so the on-board computers only fuse the navigation solutions and run guidance, control, fault detection, isolation and recovery (FDIR) and data handling.
- **Actuator commands.** An I/O driver board commands the payload hold-down and release mechanisms (HDRM), the heaters, the propulsion valves and the switched lines of the power conditioning and distribution unit (PCDU).
- **Payload data path.** Payload data are written to the on-board computers' storage (2 × 128 GB per board) and forwarded to the radio (TR-07).

5.3.2 Requirements

Table 6 lists the driving requirements of the subsystem. The complete list, with verification methods, is given in Table 14 (appendix). Requirements derive from the L1 requirements of the Technology Traceability Matrix.

Table 6: Driving Avionics & GNC requirements (TBD values to be fixed by the dispersion analysis and the system budgets).

ID	Requirement	Parent
GNC-01	The GNC subsystem shall land the Hopper within a circle of 100 m diameter (50 m radius) centred on the designated target location.	L1-03
GNC-02	The GNC subsystem shall navigate from liftoff to touchdown without ground support.	L1-03, L1-01
GNC-05	The GNC subsystem shall estimate position relative to the surface to ≤ TBD m (3σ) before the start of terminal descent, using on-board reference maps (TRN).	GNC-01
GNC-06	The GNC subsystem shall estimate altitude above ground to ≤ TBD m and velocity relative to the surface to ≤ TBD m/s (3σ) from ≤ TBD km altitude down to touchdown.	GNC-01, GNC-08
GNC-08	The GNC subsystem shall achieve touchdown with vertical velocity ≤ TBD m/s, horizontal velocity ≤ TBD m/s, tilt ≤ TBD deg and angular rates ≤ TBD deg/s.	L1-01 (safe delivery)
GNC-10	The GNC subsystem shall measure altitude with at least two dissimilar sensors (e.g. laser altimeter + radar altimeter or Doppler LIDAR range channel).	GNC-06, GNC-08
GNC-13	The GNC subsystem shall perform ≥ TBD mid-course correction manoeuvre(s) during the ballistic arc, with a total correction Δv ≤ TBD m/s.	GNC-01, L1-02
AVI-02	The avionics shall autonomously detect, isolate and recover from faults during the hop (FDIR), so that no single failure causes loss of the payload; safe-abort mode TBD.	GNC-02, L1-01
AVI-03	The avionics shall provide the payload data interface defined in the Payload ICD (RS-422 and Ethernet) at every payload position.	L1-09
AVI-05	The avionics shall store ≥ 100 GB (TBC) of payload and housekeeping data.	L1-10, L1-07
AVI-07	The avionics shall operate for ≥ 72 h at the destination during lunar day. Lunar night survival is not required.	L1-07
AVI-09	The avionics shall tolerate a total ionizing dose ≥ TBD krad and single-event effects for the mission duration without loss of function.	L1-01, L1-07
AVI-14	Every GNC sensor and actuator shall be verified end-to-end in flight configuration before launch, including interlocks, safety switches and enable plugs, under representative conditions (lighting, regolith albedo, lidar backscatter and multiple returns).	GNC-01
AVI-23	The on-board software shall execute the hop autonomously on a preloaded trajectory, from commit-to-launch to touchdown; during the hop the ground can only command abort or safe mode.	GNC-02, AVI-02
AVI-OBC-02	The on-board computer shall be implemented as TBD (baseline: 3) redundant low-cost units with majority voting.	AVI-02, AVI-09

5.3.3 Trade-offs

The main subsystem trade concerns how altitude and ground-relative velocity are sensed during descent (TR-04). It produced two candidate sensor configurations (Table 7). Both share the star trackers, the IMUs, the TRN cameras, the long-range laser altimeter and the radar altimeter. They differ in how horizontal velocity is obtained. The NASA Navigation Doppler Lidar (~15 kg) was excluded on mass.

Table 7: Navigation sensor configurations (common units: 2 star trackers, 3 IMUs, 2 TRN cameras). [a]Indicative, with placeholder inputs Δv = 3300 m/s and I_{sp} = 320 s.

	Config 1 – vision + altimeters (baseline)	**Config 2 – Doppler LIDAR**
Altitude	1 × DLEM 30 (long range) + 2 × DLEM 17 (short range, canted) + radar altimeter	1 × DLEM 30 + LUNA range channel + radar altimeter
Ground-relative velocity	Indirect: camera feature tracking fused with IMU and altimeters	Direct 3-axis measurement (LUNA)
Mass incl. margin	2.14 kg	5.16 kg (+3.02 kg, ≈ +5.6 kg propellant[a])
Power (nominal)	27.2 W	23.2 W + LUNA (not published)
Heritage	Vision-only TRN on Blue Ghost M1: touchdown 58 m from the targeted landing ellipse	LUNA not flown yet (first flight on IM-4, 2027)
Dissimilarity	Laser + radar + vision	Laser + Doppler LIDAR + radar + vision
Status	**Baseline**	Alternative. Decision once touchdown limits are set (IF-11)

The other subsystem trades are summarised in Table 8. The communication link architecture (relay vs. direct-to-Earth) is discussed in the Communications section.

Table 8: Avionics & GNC trade-off summary.

Trade	**Options**	**Decision**	**Main rationale**
Lunar navigation satellites (TR-01)	Baseline; optional aid; not considered	Optional aid	Constellations focus on the South Pole; equatorial coverage uncertain. TRN is current practice.
IMU class (TR-03)	MEMS (STIM300); FOG (LN-200S)	3 × STIM300	0.69 kg and 10.5 W less per unit. Errors handled by fusion of 3 always-on units, calibration on the surface and during coast, and thermal control.
On-board computer (TR-06)	3 COTS + voting; 2 rad-tolerant; 1 rad-hard	3 COTS + voting	Short mission → low total dose; single-event effects masked by voting; low cost. Less lunar-lander heritage than rad-tolerant boards.
TRN method (TR-05, TR-10)	Monocular crater matching; stereo; active LIDAR mapping	Monocular, 2 cameras with different FOV	Current practice (≈ 100 m precision). Stereo depth error grows with Z^2, so it is useful only in the terminal phase.
Vision processing (TR-12)	On the on-board computers; on the camera processors	On the cameras	Offloads the 3 on-board computers; the image stream stays inside the camera; two independent units.
Data network (TR-11)	Point-to-point + CAN; Ethernet; SpaceWire	Point-to-point for sensors; Ethernet backbone (TBC)	Sensors are low-rate. Cameras and payload need a high-rate link; the Q8S has Gigabit Ethernet but no SpaceWire.
Payload data path (TR-07)	Through the OBC and storage; directly to the radio	Through the OBC	Store-and-forward (≈ 25 GB per mission vs. 2 Mbps downlink); OBC enables the downlink when a link is available.

5.3.4 Equipment list

Table 9 lists the selected units for the baseline configuration. Mass margins are 5 % for flight-proven units with datasheet values and 20 % for COTS units and team estimates.

Table 9: Avionics & GNC equipment list, baseline configuration. Power with all units on; per-mode power is TBD. Harness and Communications not included. [b]Estimated by analogy (no published value).

ID	Unit	Qty	Unit mass [kg]	Mass incl. margin [kg]	Power nom. [W]	Interface
EQ-01	Star tracker – Redwire	2	0.475	1.00	3.0	RS-422
EQ-02	IMU – Safran STIM300	3	0.055	0.17	4.5	RS-422
EQ-03	On-board computer – Xiphos Q8S	3	0.064	0.23	15.0	GigE, CAN
EQ-04	OBC enclosure + I/O / voter board (estimate)	1	1.000	1.20	5.0	RS-422, discrete
EQ-05	TRN camera – Redwire Argus	2	0.140	0.29	10.0[b]	TBD (Ethernet assumed)
EQ-06	Laser altimeter, long range – Jenoptik DLEM 30	1	0.160	0.19	2.0	UART
EQ-08	Laser altimeter, short range – Jenoptik DLEM 17	2	0.025	0.06	4.0	UART
EQ-07	Radar altimeter – smartmicro	1	0.350	0.42	3.7	CAN, RS-485
Total (Config 1)				**3.57**	**47.2**	

5.3.5 Sensor use per flight phase

Table 10 maps the sensors to the hop phases of the ConOps (Section 7.1), grouped as follows:

- pre-hop on the launch site [3];
- vertical rise and ascent burn [4–5];
- ballistic coast [6];
- braking burn, split at the switch-on altitude of the laser altimeters [7];
- terminal descent below the safety descent point at 50 m [8];
- surface operations [9–11].

During the surface wait [2] the navigation sensors are off (low-power survival mode).

Initial state. The cameras look downwards for the descent, so on the surface they see only the regolith around the landing gear and cannot be matched to the crater maps. The inertial attitude comes from the star trackers. The initial position comes from the coordinates of the launch site provided by the transport service, refined with orbital imagery (LRO NAC, ≈ 0.5 m/px) and with radiometric ranging when a link is available (COM-03). With the position known, the local vertical is known as well, and the accelerometer biases can be calibrated against lunar gravity. The opposite does not work: a position estimate from star tracker and gravity vector alone is only good to the km level. The ±0.75 mg accelerometer repeatability ($\approx 7.4 \times 10^{-3}$ m/s^2) compared with lunar gravity (1.62 m/s^2) gives $\approx 4.5 \times 10^{-3}$ rad of error in the measured local vertical, i.e. ≈ 8 km on the lunar surface (R = 1737 km). The position is then corrected by the first TRN fixes during the ascent, once the cameras see enough craters, provided the camera still points at the surface during the burn (TBC).

Role of the two cameras.

- **Narrow-FOV camera:** high altitude (coast and upper braking burn), where its resolution still resolves enough craters.
- **Wide-FOV camera:** low altitude (lower braking burn and terminal descent), where the narrow field covers too small an area. In Config 1 it also provides ground-relative velocity by feature tracking, so it is a primary sensor for touchdown.
- **Heterogeneous stereo:** added only in the last hundreds of metres, since its depth error grows with Z^2 (TR-10). It supports the hazard detection required by the terminal descent.
- **Backup:** each camera processes its images independently (TR-12), so each is a degraded backup of the other.

Altimeters. The braking burn currently starts at 39 km altitude (mission analysis, TBC). This is above the estimated lunar range of all selected altimeters:

- DLEM 30: ≈ 6–8 km altitude;
- DLEM 17: ≈ 3–4 km;
- radar altimeter: 500 m.

In the upper braking burn (39 km → ≈ 6–8 km) the state is therefore estimated from camera-based TRN fused with the IMUs. Crater matching against the on-board maps gives a 3D position, including altitude above the map reference. This matches current practice: a crater-based absolute navigation system flown as a payload on Blue Ghost Mission 1 provided position measurements accurate to 100 m or better from lander imagery taken between 50 and 20 km altitude [5]. The laser altimeters take over below ≈ 6–8 km (depending on terrain albedo), where altitude above the local terrain and ground-relative velocity become critical. A longer-range unit (e.g. DLEM 45, ≈ 10–15 km) would only extend the overlap between TRN and altimetry.

Table 10: Sensor use per flight phase (preliminary, TBC); numbers refer to the ConOps phases. Upper braking: 39 km → ≈ 6–8 km; lower braking: ≈ 6–8 km → 50 m. ● used, ○ standby / health check / backup, – off. [a]Bias calibration: against lunar gravity on the surface (launch-site position known), against zero specific force in coast. [b]First TRN fixes once craters are visible; depends on camera pointing during the burn. [c]Crater fixes near the apex of the arc, TBC. [d]Feature tracking for velocity and heterogeneous stereo for hazard detection. [e]From ≈ 3–4 km (short-range lasers) and 500 m (radar). [f]Attitude for antenna pointing, if required by the link (TR-13).

Sensor	Pre-hop [3]	Rise + ascent [4–5]	Coast [6]	Braking, upper [7]	Braking, lower [7]	Terminal descent [8]	Surface [9–11]
Star trackers	●	○	●	○	○	–	○[f]
IMUs	●[a]	●	●[a]	●	●	●	–
Narrow-FOV camera	○	●[b]	●[c]	●	○	–	–
Wide-FOV camera	○	●[b]	–	○	●	●[d]	–
Laser altimeter, long range	–	–	–	–	●	●	–
Laser altimeters, short range	–	–	–	–	●[e]	●	–
Radar altimeter	–	–	–	–	●[e]	●	–

5.3.6 Budgets

Mass. 3.57 kg including unit margins for the baseline configuration (Table 9), excluding harness. Config 2 adds 3.02 kg.

Power. 47.2 W with all units on. **[Power per mode (surface wait, pre-hop, hop, surface operations) – needed by Power (IF-07).]**

Data. The reference payload (Prospect Moon) generates ≈ 25 GB over the mission. At 2 Mbps this takes ≈ 28 h of contact, i.e. contact during ≈ 40 % of the 72 h surface phase. The required storage (≥ 100 GB) is covered by the on-board computers' storage (2 × 128 GB per board), so no separate mass memory unit is needed. **[Data budget: payload interface rate (IF-12), housekeeping and flight telemetry rates, contact windows (relay vs. direct-to-Earth).]**

5.3.7 Interfaces with other subsystems

Avionics is the most interface-heavy subsystem. Table 11 summarises its interfaces. The detailed list is kept in the internal ICD.

Table 11: Avionics & GNC interfaces with other subsystems.

Subsystem	Avionics receives	Avionics provides	Status
Propulsion	Engine configuration, number of valves and transducers, throttle range and response	Valve and throttle commands; required attitude-control authority and bandwidth	Open
Mission analysis	Trajectory profile (ascent, coast, descent), visibility windows	Navigation and steering Δv losses; mid-course correction Δv	Open
Power	28 V regulated bus; switched, current-protected lines	Power per mode; line switching and telemetry. Harness ownership TBD	28 V agreed
Thermal	Number of temperature sensors and heater channels, control logic	Unit temperature ranges; closed-loop heater control	Open
Structures	Mounting, sensor fields of view, bonding, radiation shielding	FOV and mounting needs, grounding scheme, touchdown limits	Open
Payload	Payload ICD: RS-422 + Ethernet (rate TBD), 28 V, HDRM type	Data acquisition and storage; release commands	Open
Communications	Radio data and command interface	Data from storage, downlink enable, antenna pointing from the attitude estimate	Open

5.3.8 Open points and risks

- **Touchdown limits** (IF-11) decide between Config 1 and Config 2.
- **Dispersion analysis** (TR-08): propagate the navigation errors at injection to size the mid-course corrections and fix the TBD accuracy requirements.
- **Upper braking burn** (39 km → ≈ 6–8 km) relies on TRN + IMU only, with no altimeter. TRN accuracy at these altitudes and the need for a longer-range altimeter are to be checked in the dispersion analysis (see Section 5.3.5).
- **Payload data rate** (IF-12) and **HDRM type** (to be confirmed with the customer).
- **Hazard detection** in the terminal descent [8]: the baseline relies on the wide-FOV camera and heterogeneous stereo; no lidar hazard mapping is included. Performance to be assessed (hazard size, illumination).
- **Network backbone**: Ethernet vs. SpaceWire (TR-11).
- **Harness ownership** (Avionics or Power, IF-23).
- **Risk**: several units are COTS without lunar heritage (Argus, DLEM, radar altimeter). Mitigation: screening, end-to-end testing under representative conditions (AVI-14) and dissimilar redundancy.

5.4 Communications Subsystem

[antenna, communication concept]

5.5 Electrical Power Subsystem (Fabian)

[solar panels, batteries, power regulators, delivery architecture]

5.5.1 Power Budget

Table 12: Power demand per subsystem and mission phase. Phase averages, additional peak power in brackets. Last row: heaters switched off during burns.

Subsystem	Symbol	Unit	Transport	Waiting	Launch	Trajectory	Landing	Payload ops
Data handling	P_{DH}	W	10 (+5)	40 (+10)	40 (+10)	40 (+10)	40 (+10)	40 (+10)
GNC	P_{GNC}	W	0 (+0)	30 (+5)	30 (+5)	40 (+10)	150 (+30)	10 (+0)
Communications	P_{COM}	W	0 (+0)	50 (+50)	100 (+0)	100 (+0)	100 (+0)	100 (+0)
Propulsion	P_{PROP}	W	10 (+0)	20 (+30)	200 (+200)	100 (+100)	200 (+200)	10 (+0)
Thermal	P_{TH}	W	200 (+200)	200 (+200)	100 (+100)	100 (+100)	100 (+100)	50 (+50)
Payload	P_{PL}	W	20 (+10)	20 (+20)	20 (+0)	20 (+0)	20 (+0)	200 (+50)
Power	P_{PWR}	W	20 (+0)	20 (+0)	20 (+0)	20 (+0)	20 (+0)	20 (+0)
Total estimation	P_{tot}	W	260	380	510	420	630	430
Worst-case peak	$P_{\mathrm{wc,peak}}$	W	475	695	825	640	970	540

5.5.2 Subsystem Architecture

[drivers, trade-offs, system overview,]

5.6 Thermal Control Subsystem (Fabian)

[heater chooosing, insulation, radiators, temperature sensors (no chosing which ones tho)]

5.6.1 Mission Phases and Thermal Zones

[introduce the thermal zones we have and the given thermal requirements of these zones]

[introduce the thermal phases we go through and the environmetal conditions that are there during these phases]

5.6.2 Subsystem Architecture

[General intro, drivers, trade-offs, design]

5.7 Structures Subsystem (Lorenzo)

[main body (weight saving), mounting of subsystems, dampers, load bearing, shielding expect thermal shielding, material choice trades] [General intro, drivers, trade-offs, design]

5.7.1 Overview

In a lander, the structure is a critical subsystem: it must accommodate all other subsystems and protect them during launch and operations. For the hopper, two structurally critical phases are identified:

- Launch from Earth inside Starship;
- Touchdown at the end of each terminal descent.

The ascent phase of each hop is not considered structurally critical, since the accelerations and vibrations are assumed to be lower than those experienced during launch.

5.7.2 Key Drivers and Requirements

The main design driver for the structure is:

- **Survivability during the critical phases:** the structure must withstand the vibrations and accelerations of launch and the deceleration loads during landing. For the latter, a reliable energy-absorbing landing gear is essential.

The main requirements on the structure are:

- **Payload modularity:** the hopper shall accommodate multiple payloads of different sizes and masses.
- **Accommodation of all subsystems:** the structure shall provide sufficient volume, mounting interfaces and load-carrying capability for all subsystems, taking into account the size and mass of each.

The structural design follows from these drivers and requirements. The launch loads are derived from two sources. The Starship Users Guide [6] is the primary reference, since Starship is the baseline launch vehicle. However, it is a preliminary document and provides only quasi-static load factors (axial $+6/-2\,g$, lateral $\pm 2\,g$), acoustic and shock levels. In addition, Starship is currently an experimental launch vehicle, so these values are likely to change during its development. More precise values are therefore taken from the Falcon Payload User's Guide [7]. This combination is justified by the Starship guide itself, which states that Starship environments will meet or improve upon those of Falcon Heavy. The resulting load set will be revised once updated Starship data become available.

5.7.3 Main body

The main body is an open structure bounded by two hexagonal isogrid panels, which serve as the lower and upper decks of the hopper (Figure 3). All onboard subsystems are mounted on these decks, while the payload can be installed on either face of either deck. The mounting position of each payload is selected according to its dimensions and mass, as well as its operational requirements. The distance between the lower deck and the lunar surface is $60\,\mathrm{cm}$, which allows payloads to interact directly with the regolith and enables the deployment of small rovers. This configuration was chosen with modularity as a priority.

Both decks are made of aluminum–lithium alloy 2195 (Al–Li 2195) and are connected by a truss structure of carbon-fiber-reinforced polymer (CFRP). These materials were selected primarily to minimize the structural mass of the spacecraft. In addition, both materials have extensive flight heritage, wich reduces the qualification cost.

Each deck features a standardized grid of threaded inserts for M(TBD) fasteners, through which subsystems and payloads are attached to the primary structure. This solution is inherited from the payload mounting decks of Astrobotic's Peregrine lander [8].



Figure 3: Preliminary CAD model of the Hopper's Structure

5.7.4 Landing Gear

The landing gear is an essential part of the structure: it ensures stability on sloped terrain and absorbs the impact loads at touchdown. The landing energy is absorbed by crushable aluminum honeycomb, which dissipates it through plastic deformation. This technology has a long flight heritage, dating back to the Apollo missions [9].

When loaded along its cell axis, pre-crushed aluminum honeycomb deforms plastically by progressive folding of the cell walls at an almost constant crush stress σ_c, up to about $70\,\%$ of its length [10]. The axial force in each leg is therefore limited to $F_c = \sigma_c \times A$, and the energy absorbed over a stroke s is simply $E = F_c \times s$. Unlike a viscous damper, the honeycomb produces no rebound and its force does not depend on the touchdown velocity, which makes its behavior highly predictable.

This solution was selected for the following reasons:

- **Low density:** aluminum honeycomb is very lightweight, with densities between 16 and $150\,\mathrm{kg/m^3}$ [10].
- **Energy-absorption efficiency:** it reaches a specific energy absorption of up to $30\,\mathrm{kJ/kg}$ [10], making it one of the most mass-efficient flight-proven landing absorbers.

The main drawback of this energy absorber is that its deformation is irreversible. This is consistent with the single-hop mission strategy, but it limits the flexibility of the mission, since additional hops would require a reusable damper. A further drawback is the sensitivity of the crush strength to temperature; the absorber is therefore protected by thermal insulation, as on the Apollo Lunar Module [9]. **[Detalied trade study is needed once the STR mass is refined]**

Chapter 6

Payload Integration Subsystem (Lorenzo)

[payload description, design, adaptible payload interface]

6.1 Payload Interface

6.1.1 Overview

This section describes the payload interfaces of the hopper. The Spacecraft is designed to be payload-agnostic: its mechanical, electrical and data interfaces are sized for the Interlune's payload as the reference case, but can also accommodate other payloads within the same envelope.

6.1.2 Mechanical Interface

The mechanical interface between the payloads and the hopper is provided by the regular bolt pattern of the two decks. Each payload is assigned a dedicated set of threaded inserts and is fastened to them either directly or through an adapter plate, if its mounting footprint does not match the grid. In this way, the design of each payload is independent of the hopper structure, and different payloads can be integrated without modifications to the primary structure. This ensures a robust and secure fixing of the payloads while keeping the platform modular.

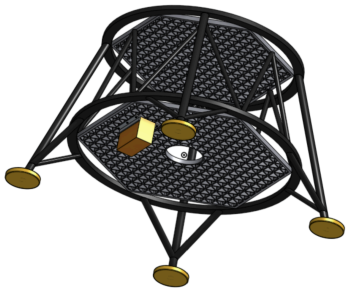


Figure 4: Payload position in the Hopper

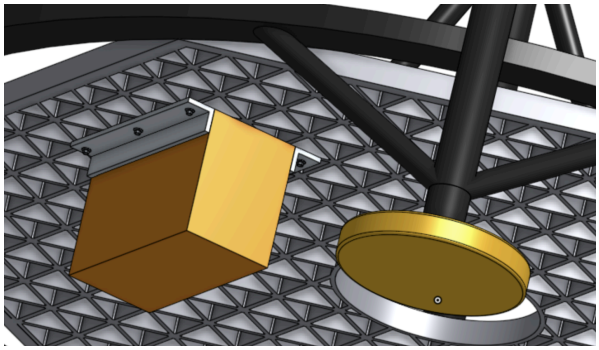


Figure 5: Payload mechanical interface: threaded-insert grid on the lower deck and adapter plate for payload mounting.

6.1.3 Electrical & Data Interface

Each payload is supplied through a dedicated power line and a dedicated data line, both terminated with standardized connectors. Power is provided on a $28\,\mathrm{V}$ bus, the standard supply voltage for spacecraft payloads, with a maximum of $300\,\mathrm{W}$ as specified by the industrial partner. Payload data are transferred

over a shielded Ethernet, which maximizes compatibility between the payload and the onboard computer. An additional RS-422 line is available, if required, for low-rate command and housekeeping communication between the avionics and the payload.

6.2 Prospect Moon payload

6.2.1 Overview

The primary payload of the Prospect Moon mission is described below. The mission aims to identify the resources contained in lunar regolith and to assess what is required to extract them.

The payload occupies an envelope of $33.02 \times 19.05 \times 25.40\,\mathrm{cm}$ and carries an $80\,\mathrm{cm}$ robotic arm, which is folded at mid-length during flight. There is also a mass spectrometer which has the following dimension $15.5 \times 20 \times 46\,\mathrm{cm}$. The total payload mass is approximately $45\,\mathrm{kg}$.

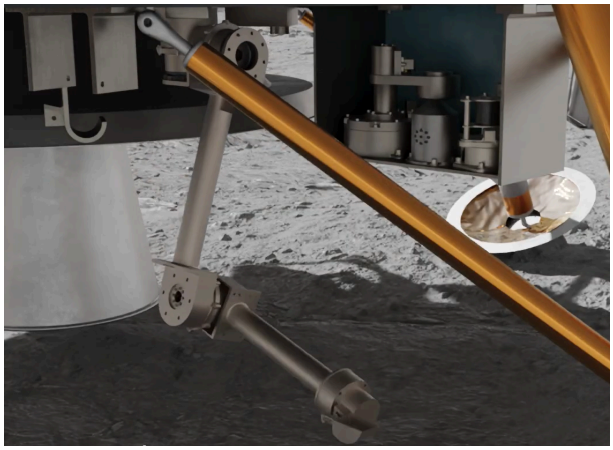


Figure 6: Rendering of the Prospect Moon payload [11].

6.2.2 Instrument Design

The Payload will have 3 main instrument inside: a device to mechanically process a regolith sample via milling; a regolith-sorting device; a multi-spectral camera; and a mass spectrometer.

6.2.3 Measurement Principle

The robotic arm and scoop will collect Three samples of regolith:

- The first scoop will be heated to release volatile gases from the regolith, and the mass spectrometer will measure their concentrations.
- The second scoop will test mechanical processing technology to release gases, and the mass spectrometer will again measure concentrations.
- The third scoop will test Interlune's technology to sort lunar regolith by grain size, a separate step in the harvesting system.

By comparing gas concentration from both thermal and mechanical process, Interlune will validate that the second method is able to release gases with enough concentration to be harvested.

For each scoop a multispectral camera will flash at multiple wavelengths to collect data on titanium levels and regolith ages. With these data is possible to predict the amount of Solar gases, including helium-3, present in the regolith.

Chapter 7

Mission Operations and Ground Segment (Freddy)

7.1 Concept of Operations

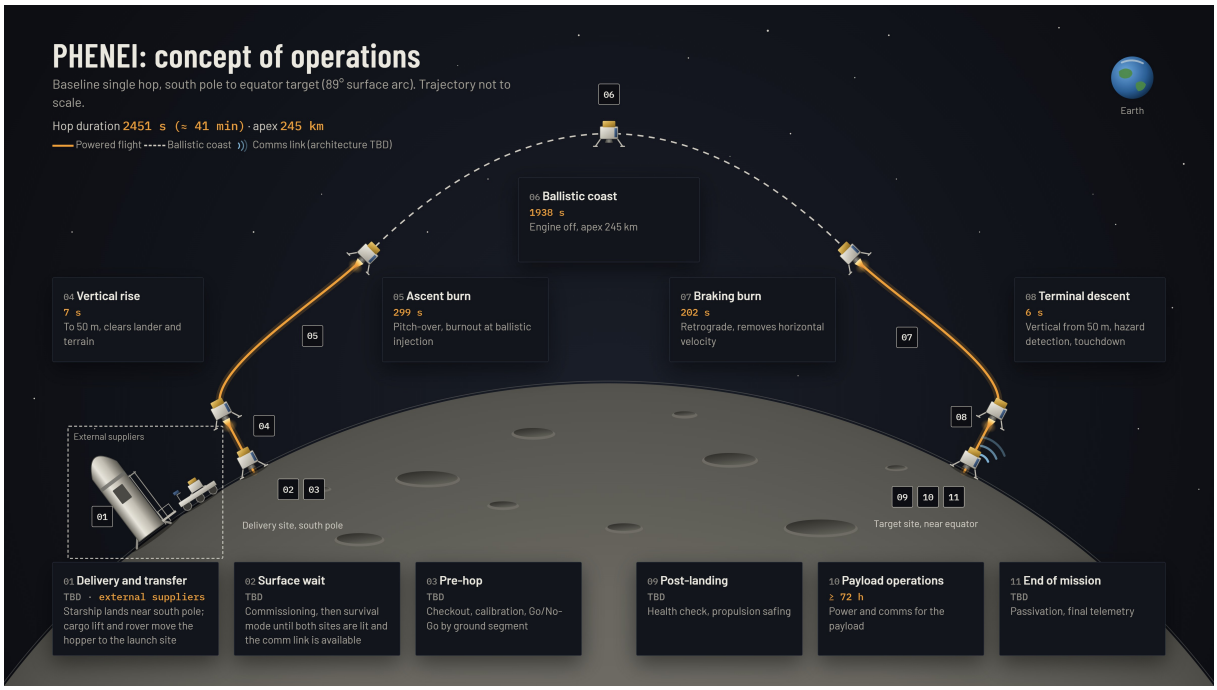


Figure 7: Preliminary ConOps

The concept of operations comprises phases owned by the PHENEI mission team and phases provided by external suppliers. Numbers in brackets refer to the ConOps figure.

A. Arrival and mission pre-phase

The Lunar hopper arrives on the lunar surface aboard a commercial heavy lander such as the lunar Starship.

- **[1] Delivery and transfer** *(external suppliers)*
  - Unloading from Starship (anticipated via elevator) and short-distance transport to a suitable launch site are provided by third parties (SpaceX or others).
  - After arrival at the launch site and unloading from the transport vehicle, PHENEI takes over.
- **[2] Surface wait**
  - After handover, the hopper is commissioned: first contact with the ground segment is established and a health check of all subsystems is performed.

  - The hopper then remains at the launch site until a suitable launch window opens. The length of this waiting period depends on the target site, since the hop has to be timed such that the target site is illuminated at arrival and during the subsequent payload operations.
  - During the wait, the hopper operates in a low-power survival mode. This requires a launch site with near-continuous illumination, which is only available at selected elevated locations near the south pole. The launch site therefore has to fulfill specific illumination requirements **[Illumination requirements for the launch site and selection of suitable locations.]**.
- **[3] Pre-hop**
  - When the optimal window arrives **[Definition of the "optimal window"; preliminary: maximize illuminated time at the target site.]**, the hopper performs automated checks.
  - It then waits for a Go/No-Go decision by the ground segment, based on the required flight-readiness conditions.

B. Hop phase

- **[4] Vertical rise:** On the Go signal, the hopper autonomously performs a vertical takeoff to a safety height of **[Safety height; preliminary 50 m, further analysis planned.]**.
- **[5] Ascent burn:** Once at the safety height, the hopper autonomously performs the ascent burn, adjusting spacecraft pitch and the resulting total thrust angle.
- **[6] Ballistic coast:** At the ballistic injection point, the hopper turns off its main thrusters and coasts on a ballistic trajectory over the lunar surface.
- **[7] Braking burn:** For descent initiation, the hopper autonomously performs a braking maneuver until it reaches the safety descent point **[Safety descent point; preliminary 50 m altitude, no horizontal velocity except for hazard avoidance.]**.
- **[8] Terminal descent:** From the safety descent point, the hopper autonomously performs a vertical descent with active hazard avoidance, detecting hazards and maneuvering around them until touch-down **[Hazard avoidance is not included in the phase durations of the ConOps figure.]**.

C. Surface operations

- **[9] Post-landing**
  - Post-landing checks and propulsion safing.
  - Transfer of health data to Earth.
- **[10] Payload operations**
  - The hopper supports the payload with an average power of 200 W and a downlink data rate of 2 Mbps to Earth.
  - Nominal mission duration: at least 72 h.
  - If the hopper remains operational beyond this, payload operations may continue until the available power no longer supports them.
- **[11] End of mission**
  - End of mission is declared once payload operations can no longer be sustained.
  - The hopper is then passivated (remaining energy sources safed, transmitters switched off) and remains at the landing site.

7.2 Hop Trajectories

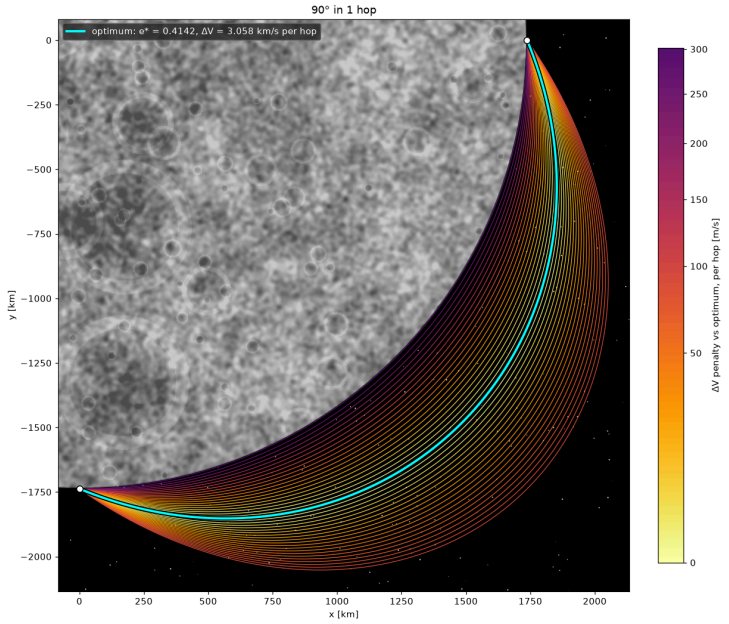


Figure 8: Analytic trajectory optimization

Impulsive reference hop

A common lower bound for a ballistic surface-to-surface transfer with zero velocity at both ends is a two-impulse Keplerian arc connecting the launch and target sites.**[ressource will be given]** This closed-form solution serves as the baseline: among all Keplerian arcs passing through both points, the one requiring the minimum total Δv is selected (minimum-energy trajectory). Both burns, one at liftoff and one at touchdown, are assumed to be impulsive, i.e. with infinite thrust, so no gravity or steering losses occur (Figure 8).

For a single hop from the lunar south pole to the equator (central angle $89°$, i.e. a quarter of the lunar circumference, approx. 2730 km), the baseline values are:

- Δv: 3050 m/s
- Flight time: 41.1 min
- Apoapsis altitude: 360 km

While the closed-form solution gives a good first estimate of the order of magnitude of the Δv budget, it is limited to the idealized case of a two-impulse ballistic hop. It does not represent the trajectory planned in the ConOps, which includes finite thrust and vertical takeoff and landing phases.

Numerical trajectory optimization

To capture finite-thrust effects, the trajectory is optimized numerically using a planar point-mass model in a Moon-centered, inertial polar coordinate system (r, θ). The plane of motion is the great-circle plane through the launch and target sites.

The state and input vectors are

$$\boldsymbol{x} = \left(r, \theta, \dot{r}, \dot{\theta}, m\right)^T, \quad \boldsymbol{u} = (T, \beta)^T \tag{1}$$

with radial position r, range angle θ, radial velocity $\dot{r}$, angular rate $\dot{\theta}$ and vehicle mass m. The thrust magnitude is bounded to $T \in [0, T_{\max}]$ **[Maximum thrust $T_{\max}$ still to be determined; 6 kN assumed in**

this version.], and the thrust angle β is measured from the local horizontal ($\beta = 0$: along $+\theta$, $\beta = \pi/2$: radially outward).

The equations of motion are

$$\dot{\boldsymbol{x}} = \begin{pmatrix} \dot{r} \\ \dot{\theta} \\ T/m \sin\beta + r\dot{\theta}^2 - \mu/r^2 \\ \left(T/m \cos\beta - 2\dot{r}\dot{\theta}\right)/r \\ -T/\left(I_{\mathrm{sp}} g_0\right) \end{pmatrix} \tag{2}$$

where μ is the lunar gravitational parameter and g_0 the standard gravitational acceleration.

The model is based on the following assumptions:

- **Non-rotating Moon:** the surface velocity due to lunar rotation is at most approx. 4.6 m/s at the equator and zero at the pole, i.e. below 0.2 % of the hop Δv.
- **Point-mass gravity field:** the Moon is treated as a homogeneous sphere, gravity anomalies (mascons) are neglected.
- **Planar motion:** no out-of-plane motion.
- **No third-body perturbations:** Earth and Sun are neglected over the hop duration of approx. 41 min.
- **No attitude or throttle dynamics:** thrust magnitude and direction can change instantaneously **[Rate limits on attitude and throttle, to be implemented for the PDR.]**.
- **Spherical surface:** no terrain or craters.
- **Constant I_{sp}:** mass flow is proportional to thrust **[Throttle-dependent propellant consumption model, to be implemented for the PDR.]**.

Optimization Problem

The optimization is conducted in two stages. A single-phase numerical optimization provides trajectory data as a warm start for the second stage, a multiphase numerical optimization. The objective is solely minimal fuel consumption, i.e. minimizing the initial mass, while the final mass is fixed to the dry mass (dry mass is governed by the structure and assumed to be given, no residual propellant is considered). The problem is formulated in CasADi [12], a framework for nonlinear optimization with automatic differentiation, and solved with the interior-point solver IPOPT **[citation needed?]**. The dynamics are discretized with one RK4 step per interval, piecewise constant controls and free final time. Since the problem is nonconvex, the solver requires a good initial guess for the trajectories. For the single-phase problem, a simple analytic guess is sufficient, whereas the multiphase problem needs a guess that is consistent with the phase boundaries and event constraints. That is the reason for the two-stage optimization: the single-phase optimum serves as the initial guess for the multiphase optimization, resulting in faster and more reliable convergence. Boundary conditions, constraints and initial guesses can be found in the appendix(Appendix D).

Simulation results

The multiphase optimization converged to a solution that is consistent with the verification checks. Table 13 compares the Δv of the impulsive baseline and the multiphase solution, and Figure 9 shows the optimized trajectory. For the hop from 89° S to the equator, the multiphase solution requires 6.3 % more Δv than the impulsive baseline. More specifically, the losses consist of 182.2 m/s gravity loss and 77.3 m/s steering loss. Their sum exceeds the Δv difference to the impulsive solution, since burnout

occurs at altitude and therefore requires a lower burnout velocity than the impulsive surface burn. The total propellant mass required for a dry mass of $m_{\text{dry}} = 500$ kg is $m_{\text{prop}} = 907.6$ kg.

Table 13: Total Δv, flight time and losses of the analytic and multi-phase hop solution for an 89° transfer.

Method	**Δv (m/s)**	**vs. impulsive**	**Flight time (min)**	**Gravity loss (m/s)**	**Steering loss (m/s)**
Impulsive (analytic)	3049.7	–	41.1	0	0
Multiphase	3240.6	+6.3 %	40.9	182.2	77.3

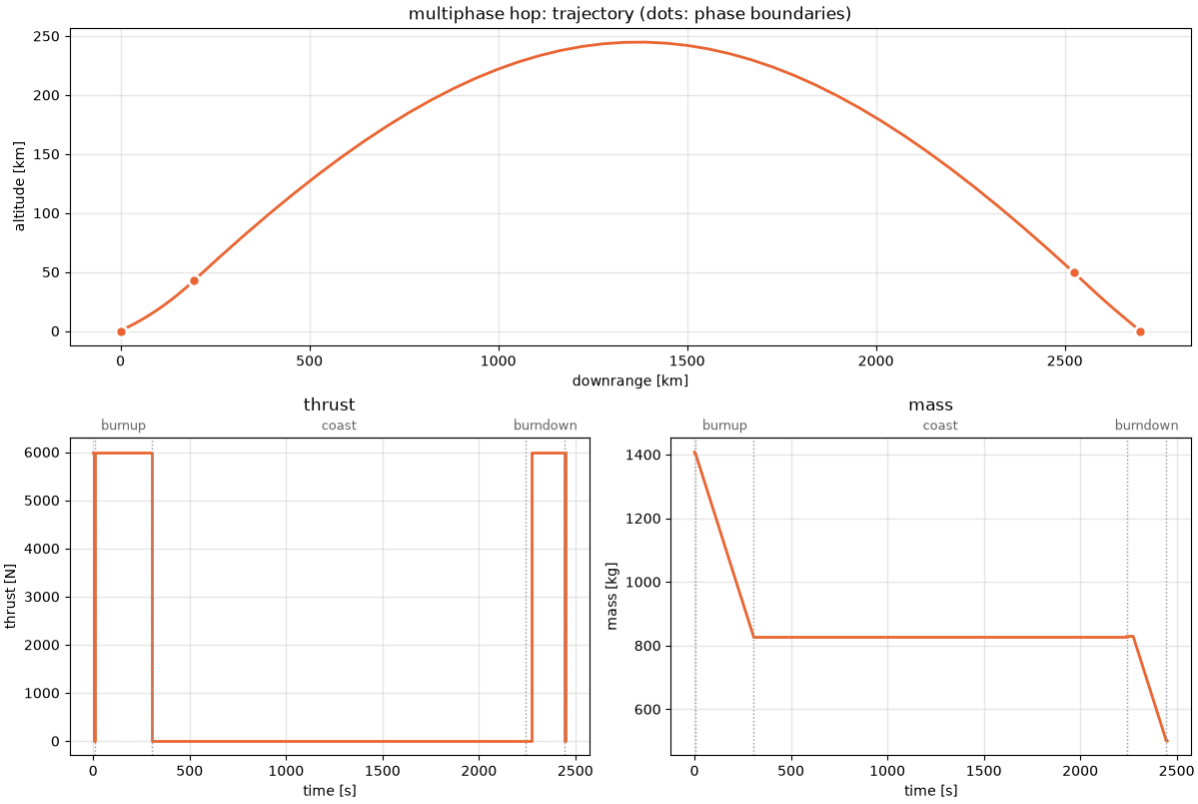


Figure 9: Optimized multi-phase trajectory for the 89° hop.

[Further discussion of the trajectory planned, also maps of distance vs mass consumption since it's essential for the economical use of the mission.]

7.3 Ground Segment

[Ground segment architecture (ground stations, mission control) to be defined.]

Chapter 8

Management

8.1 Project Management Plan

[how is the team structured an stuff]

8.1.1 Schedule

The programs schedule follows the NASA's Project Life Cycle given in SMAD [2]. Choosing this schedule is reasonable, since all our stakeholders are based in the USA and partly funded by NASA **[reference]**.

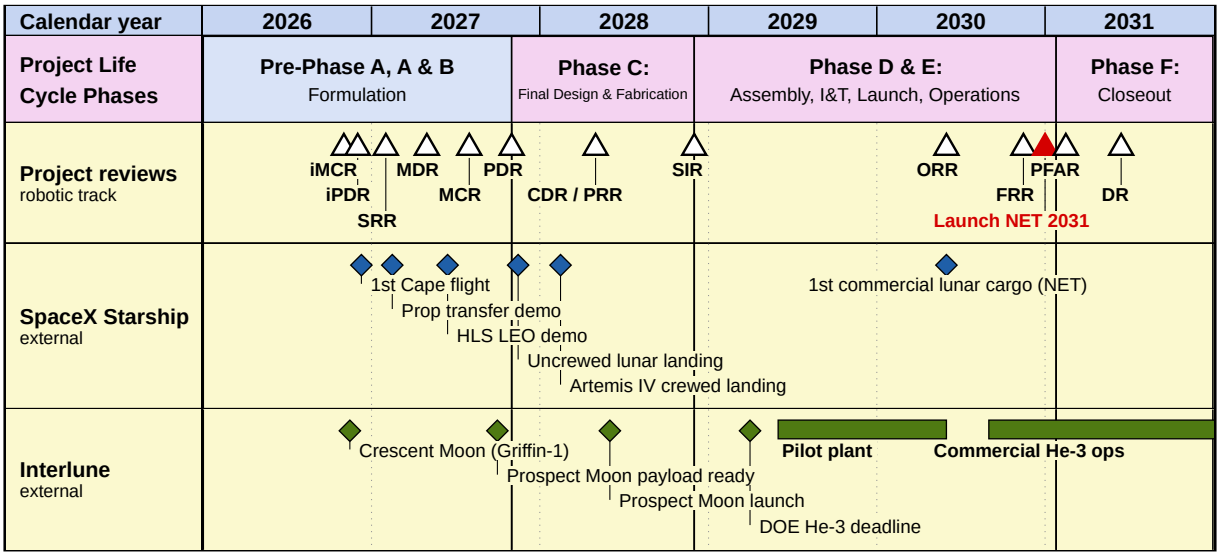


Figure 10: Top-level programme schedule following the NASA project life cycle, with SpaceX Starship and Interlune milestones.

Pre-Phase A, Phase A & B

The Formulation Phases are started by PHENEI withing the scope of the Mission Project of the Space Systems Master at ETH Zürich. A process that normally takes around two years **[give source for that]** is compressed within 10 weeks, Starting with the kick-off in late September 2026 and ending with the internal PDR (iPDR) and project handover to Interlune in early December 2026.

Starting from Interlune's L1 requirements **[maybe give a reference for that]** PHENEI finalizes the mission objectives and requirements, performs key trade studies and begins with the preliminary designs of the lunar hopper's subsystems. After the handover, the formulation continues in 2027 with the System Requirements Review (SRR), the Mission Definition Review (MDR) and the Mission Concept Review (MCR), and ends with the PDR in late 2027.

Phase C

Phase C starts from the PDR in late 2027 and ends with the system integration review at the end of 2028. During that Phase the lunar hopper's design in completed and confirmed with the CDR in mid 2028 after which flight hardware fabrication begins. Interlune plans to have its Prospect Moon payload ready for integration in autumn 2027 with its first flight with an other lander in 2028. As the lunar hoppers first mission is to integrate the Prospect Moon payload into the spacecraft the interface with interlune is already fixed before the CDR. The Results from the first flight in 2028 can feed back into the design of the hopper.

Already during the formulation, SpaceX plans orbital propellant transfer in early 2027 and and uncrewed Starship lunar landing by the end of 2027. These demonstrations are a main external milestone for our schedule, since their outcomes show if a commercial Starship cargo flight in 2030 is realistic.

Phase D & E

Phase D starts with the system integration review at the end of 2028. The subsystems are assembled into a complete hopper and the Prospect Moon payload is integrated. The flight system and the payload will go through a extensive testing phase including environmental testing. After the the operational readiness review in mid 2030 the hopper is transported to the launch site and integrated into the Starship cargo, where it will undergo the flight readiness review. The launch date will be set by SpaceX and not by the Project and is currently planned for no earlier than 2031. The project deliberately does not plan to fly on the first commercial Starship lunar cargo flight (NET 2030), but on a later one. This gives margin, since the Starship's lunar programm has slipped alread several times **[give a reference here]**.

Phase E begins with the launch of the Starship. It will be refueled in low earth orbit and eventually lands on the Shackleton connecting ridge. After offloading the hopper can start with its operation of hoping to its target site on the lunar equator and handing over to Interlune. The whole operation will take around two to three earth days.

Phase F

Once all mission and payload data have been downlinked, the hopper is passivated and left at its landing site; no recovery or disposal is planned. The flight data and lessons learned are then assessed and used to adapt the hopper design for new customers, starting the next iteration of the life cycle.

8.2 Cost

[TBD]

8.3 Life Cycle Assessment

[Full data.]

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Appendix A

Product Tree

[TBD]

Appendix B

Propulsion

[TODO: improve citations to SMAD adding the precise equation, check whether it's possible to improve the current model, rework engine mass estimation (fine for a very rough estimate), but it has to be grounded in literature material / empirical data]

B.1 Notation

Subscripts: oxi oxidiser (MON-3), fuel fuel (MMH), prop propellant (oxidiser + fuel), tank propellant tank, pres pressurant (He) system, eq equivalent. Subscript $i \in \{\mathrm{oxi}, \mathrm{fuel}\}$ stands for either propellant.

Symbol	Meaning	Unit
m_{prop}	loaded propellant mass (model input)	kg
$m_{\mathrm{oxi}}, m_{\mathrm{fuel}}$	oxidiser and fuel mass	kg
R_m	mixture ratio $m_{\mathrm{oxi}}/m_{\mathrm{fuel}}$	–
$\rho_{\mathrm{oxi}}, \rho_{\mathrm{fuel}}$	propellant densities at the hot case	kg/m^3
ε	ullage fraction	–
$V_{\mathrm{tank,oxi}}, V_{\mathrm{tank,fuel}}$	tank volumes	m^3
c_j	SMAD tank fit coefficients (V in litres)	kg/Lj
γ_j	SMAD COPV fit coefficients (X in MPa m^3)	kg/(MPa m^3)j
$P_{u,i}$	pressure required at the engine inlet	Pa
$\Delta P_{u,i}$	margin between tank gas and engine inlet	Pa
ΔP_{reg}	minimum bottle-to-tank regulator differential	Pa
$P_{0,\mathrm{pres}}$	bottle pressure when full	Pa
$P_{f,\mathrm{pres}}$	bottle pressure when “empty”	Pa
T_i	tank gas temperature, cold case	K
T_{pres}	bottle temperature at which $P_{0,\mathrm{pres}}$ is reached	K
M_{pres}	molar mass of the pressurant (He)	kg/mol
R	universal gas constant	J/(mol K)
n_{pres}	moles of gas delivered to the tanks	mol
n^*_{pres}	moles of gas loaded in the bottle (includes residual)	mol
V_{pres}	bottle volume	m^3
X	bottle size parameter $P_{0,\mathrm{pres}} V_{\mathrm{pres}}$	MPa m^3

Equivalent quantities, each defined where it appears: $\rho_{\mathrm{eq,prop}}$ (Equation 10), $M_{\mathrm{eq,prop}}$ (Equation 20), $E_{\mathrm{eq,pres}}$ and $\rho_{\mathrm{eq,pres}}$ (Equation 22), R_p (Equation 21). Each one maps the propellant mass m_{prop} to a quantity of the same kind of a single fictitious propellant, so the whole system reads as one tank and one bottle.

B.2 Masses

From $R_m = m_{\text{oxi}}/m_{\text{fuel}}$ and $m_{\text{prop}} = m_{\text{oxi}} + m_{\text{fuel}}$ we get $m_{\text{prop}} = (1 + R_m)m_{\text{fuel}}$, hence

$$m_{\text{oxi}} = \frac{R_m}{1 + R_m}\, m_{\text{prop}}, \quad m_{\text{fuel}} = \frac{1}{1 + R_m}\, m_{\text{prop}}. \tag{3}$$

B.3 Densities

Fits with T in kelvin and ρ in kg/m³ **[ADD CITATION FOR NASA PAPER in which I found them]**:

$$\rho_{\text{MON-0}}(T) = 2066 - 1.979\,T - 4.826 \times 10^{-4}\,T^2, \tag{4}$$

$$\rho_{\text{MON-10}}(T) = 1582.5 + 1.280\,T - 6.26 \times 10^{-3}\,T^2. \tag{5}$$

MON-k contains $k\%$ of NO and $(100 - k)\%$ of N_2O_4. Writing the density as a linear blend of the two reference mixtures, $\rho_{\text{MON-k}} = A\rho_{\text{MON-0}} + B\rho_{\text{MON-10}}$ with $A + B = 1$ and $100 - k = 100A + 90B$, gives $B = k/10$ and

$$\rho_{\text{oxi}}(T) = \left(1 - \frac{k}{10}\right) \rho_{\text{MON-0}}(T) + \frac{k}{10}\, \rho_{\text{MON-10}}(T). \tag{6}$$

$$\rho_{\text{fuel}}(T) = 1150.34 - 0.93949\,T. \tag{7}$$

Both are evaluated at the hot case, where the density is lowest, so the volumes are largest.

B.4 Tank volumes

A tank holds the liquid volume m_i/ρ_i plus the ullage fraction ε:

$$V_{\text{tank,oxi}} = (1 + \varepsilon)\, \frac{R_m}{(1 + R_m)\, \rho_{\text{oxi}}}\, m_{\text{prop}}, \quad V_{\text{tank,fuel}} = (1 + \varepsilon)\, \frac{1}{(1 + R_m)\, \rho_{\text{fuel}}}\, m_{\text{prop}}. \tag{8}$$

The oxidiser section carries the factor R_m because it holds $R_m/(1 + R_m)$ of the propellant. The propellant is stored in one bi-propellant tank made of an oxidiser and a fuel section, so the tank volume is the sum

$$V_{\text{tank,prop}} = V_{\text{tank,oxi}} + V_{\text{tank,fuel}} = (1 + \varepsilon)\, \frac{m_{\text{prop}}}{\rho_{\text{eq,prop}}}, \tag{9}$$

which defines the equivalent propellant density

$$\frac{1}{\rho_{\text{eq,prop}}} = \frac{1}{1 + R_m} \left(\frac{R_m}{\rho_{\text{oxi}}} + \frac{1}{\rho_{\text{fuel}}} \right), \quad \rho_{\text{eq,prop}} = \frac{(1 + R_m)\, \rho_{\text{oxi}}\, \rho_{\text{fuel}}}{\rho_{\text{oxi}} + R_m\, \rho_{\text{fuel}}}. \tag{10}$$

It is the density of one fictitious liquid that fills the same total volume as the two real ones. It is exactly what the tank mass fit needs: one volume in, one mass out. This is valid only because a bi-propellant tank is used. $V_{\text{tank,oxi}}$ and $V_{\text{tank,fuel}}$ are still needed separately later, because the two sections sit at different temperatures and pressures and set the moles of pressurant.

B.5 Propellant tank mass

The bi-propellant **[ADD REFERENCE TO CATALOGUE]** tank follows the SMAD [2] pressure-fed (PMD) fit, with the volume in litres:

$$m_{\text{tank,prop}}(V) = c_3 V^3 + c_2 V^2 + c_1 V + c_0. \tag{11}$$

With the single equivalent volume $V_{\text{tank,prop}} = (1 + \varepsilon) m_{\text{prop}}/\rho_{\text{eq,prop}}$ and 1 m³ = 1000 L, the volume in litres per kg of propellant is

$$a_{\text{tank,prop}} = 1000\frac{1+\varepsilon}{\rho_{\text{eq,prop}}}, \quad V_{\text{tank,prop}}[\text{L}] = a_{\text{tank,prop}}\, m_{\text{prop}}. \tag{12}$$

Substituting in Equation 11 gives a cubic in m_{prop}:

$$m_{\text{tank,prop}} = k_{3,\text{tank}}\, m_{\text{prop}}^3 + k_{2,\text{tank}}\, m_{\text{prop}}^2 + k_{1,\text{tank}}\, m_{\text{prop}} + k_{0,\text{tank}}, \tag{13}$$

$$k_{j,\text{tank}} = c_j\, a_{\text{tank,prop}}^j \quad (j = 1, 2, 3), \quad k_{0,\text{tank}} = c_0. \tag{14}$$

B.6 Pressure convention

Three different pressures, with three different symbols. The engine needs $P_{u,i}$ at its inlet; the tank gas must be higher by $\Delta P_{u,i}$ so that the flow never reverses and lines and regulator losses are covered. The regulator keeps this tank gas pressure constant:

$$P_{\text{tank},\,i} = P_{u,i} + \Delta P_{u,i}. \tag{15}$$

A single bottle feeds both sections of the tank. When it is nearly empty it must still exceed the higher tank pressure by the regulator differential ΔP_{reg}:

$$P_{f,\text{pres}} = \max\big(P_{\text{tank,oxi}}, P_{\text{tank,fuel}}\big) + \Delta P_{\text{reg}}. \tag{16}$$

The bottle starts at $P_{0,\text{pres}}$ and ends at $P_{f,\text{pres}}$.

B.7 Gas to be delivered

As the liquid leaves, the gas fills the volume it vacates, so at the end of the burn it has filled each tank volume at the tank pressure. The ideal gas law at the cold-case temperature T_i (cold gas is densest and needs the most moles) gives the moles delivered to each tank, and the sum with Equation 8 gives

$$n_{\text{pres}} = \sum_i \frac{P_{\text{tank},i} V_{\text{tank},i}}{R T_i} = \frac{m_{\text{prop}}}{R}\,\frac{1+\varepsilon}{1+R_m}\left[\frac{P_{\text{tank,oxi}} R_m}{\rho_{\text{oxi}} T_{\text{oxi}}} + \frac{P_{\text{tank,fuel}}}{\rho_{\text{fuel}} T_{\text{fuel}}}\right]. \tag{17}$$

The quantity is linear in m_{prop}. The delivered moles are not the loaded moles, see the next step.

B.8 Residual gas

The regulator stops working below $P_{f,\text{pres}}$, so gas stays in the bottle. Let the bottle be filled with n^*_{pres} moles at $P_{0,\text{pres}}$, temperature T_{pres} and volume V_{pres}:

$$P_{0,\text{pres}} V_{\text{pres}} = n^*_{\text{pres}} R T_{\text{pres}}. \tag{18}$$

At the end of the burn the gas left is $n_r = P_{f,\text{pres}} V_{\text{pres}}/(RT_{\text{pres}}) = n^*_{\text{pres}} P_{f,\text{pres}}/P_{0,\text{pres}}$, using Equation 18. The delivered amount is $n_{\text{pres}} = n^*_{\text{pres}} - n_r$, so

$$n^*_{\text{pres}} = n_{\text{pres}}\,\frac{P_{0,\text{pres}}}{P_{0,\text{pres}} - P_{f,\text{pres}}} = f_{\text{res}}\, n_{\text{pres}}, \quad f_{\text{res}} = \frac{1}{1 - P_{f,\text{pres}}/P_{0,\text{pres}}} > 1. \tag{19}$$

This correction applies to the pressurant gas mass and to the bottle size.

B.9 Equivalent molar mass

Define the propellant mass per mole of loaded gas, $M_{\text{eq,prop}} = m_{\text{prop}}/n^*_{\text{pres}}$. Using Equation 17 and Equation 19:

$$\frac{1}{M_{\text{eq,prop}}} = \frac{n^*_{\text{pres}}}{m_{\text{prop}}} = \frac{f_{\text{res}}}{R}\,\frac{1+\varepsilon}{1+R_m}\left[\frac{P_{\text{tank,oxi}}R_m}{\rho_{\text{oxi}}T_{\text{oxi}}} + \frac{P_{\text{tank,fuel}}}{\rho_{\text{fuel}}T_{\text{fuel}}}\right]. \tag{20}$$

Despite the name, $M_{\text{eq,prop}}$ is not a molar mass of a substance: it is how many kilograms of propellant are expelled per mole of gas loaded, in kg/mol.

B.10 Pressurant gas mass

$$m_{\text{pres}} = M_{\text{pres}}\, n^*_{\text{pres}} = R_p\, m_{\text{prop}}, \quad R_p = \frac{M_{\text{pres}}}{M_{\text{eq,prop}}}. \tag{21}$$

R_p is the pressurisation ratio, in kg of gas per kg of propellant. The gas is inert and counts as dry mass.

B.11 Bottle volume and size parameter

From Equation 18 and the definition of $M_{\text{eq,prop}}$,

$$P_{0,\text{pres}}V_{\text{pres}} = n^*_{\text{pres}}RT_{\text{pres}} = m_{\text{prop}}\,E_{\text{eq,pres}}, \quad E_{\text{eq,pres}} = \frac{RT_{\text{pres}}}{M_{\text{eq,prop}}}. \tag{22}$$

$E_{\text{eq,pres}}$ is the pressure-volume energy of the gas per kg of propellant, in J/kg. Dividing by the full pressure gives the bottle volume, and defines the equivalent density

$$V_{\text{pres}} = \frac{m_{\text{prop}}}{\rho_{\text{eq,pres}}}, \quad \rho_{\text{eq,pres}} = \frac{P_{0,\text{pres}}}{E_{\text{eq,pres}}} = \frac{P_{0,\text{pres}}M_{\text{eq,prop}}}{RT_{\text{pres}}}. \tag{23}$$

Here $\rho_{\text{eq,pres}}$ is the propellant mass served per unit bottle volume (in kg/m³); it plays for the bottle the role that $\rho_{\text{eq,prop}}$ plays for the tanks.

B.12 Bottle mass

The SMAD [2] COPV fit uses $X = P_{0,\text{pres}}V_{\text{pres}}$ in MPa m³. Converting J to MPa m³ with 10^{-6}:

$$X = 10^{-6}\,E_{\text{eq,pres}}\,m_{\text{prop}}. \tag{24}$$

$$m_{\text{tank,pres}}(X) = \gamma_2 X^2 + \gamma_1 X + \gamma_0, \tag{25}$$

valid for $X \geq 1$ MPa m³. With Equation 24:

$$m_{\text{tank,pres}} = k_{2,\text{pres}}\,m_{\text{prop}}^2 + k_{1,\text{pres}}\,m_{\text{prop}} + k_{0,\text{pres}}, \tag{26}$$

$$k_{2,\text{pres}} = \gamma_2\left(10^{-6}E_{\text{eq,pres}}\right)^2, \quad k_{1,\text{pres}} = \gamma_1\,10^{-6}E_{\text{eq,pres}}, \quad k_{0,\text{pres}} = \gamma_0. \tag{27}$$

The validity limit translates into a minimum propellant mass $m_{\text{prop}} \geq 10^6/E_{\text{eq,pres}}$.

B.13 Assembly

The propellant-dependent dry mass is the sum of tanks, bottle and gas:

$$m_{\text{sys,prop}}(m_{\text{prop}}) = m_{\text{tank,prop}} + m_{\text{tank,pres}} + m_{\text{pres}} = K_3\,m_{\text{prop}}^3 + K_2\,m_{\text{prop}}^2 + K_1\,m_{\text{prop}} + K_0, \tag{28}$$

$$K_3 = k_{3,\text{tank}}, \quad K_2 = k_{2,\text{tank}} + k_{2,\text{pres}}, \quad K_1 = k_{1,\text{tank}} + k_{1,\text{pres}} + R_p, \quad K_0 = k_{0,\text{tank}} + k_{0,\text{pres}}. \tag{29}$$

The total volume to accommodate is $V_{\text{tank,prop}} + V_{\text{pres}}$.

[Mass required by the ACS is not modelled yet]

B.14 Engines (first estimate)

For a total thrust T and reference engine $(T_\mathrm{b}, m_\mathrm{b})$, with $x = \max(T, T_\mathrm{b})/T_\mathrm{b}$:

[The idea is that of having a commercially-available engine to work off-design to accommodate thrust needs between integer multiples of mass and thrust characteristics. The function is continuous so that it can be optimised for in the trajectory optimisation; it penalises thrust requirements far from its nominal range while tolerating small deltas. However, if I'm not able to justify this approach with literature support, I will rework it. Literature proposes sublinear mass scaling with thrust, but that relates to designing a new engine from scratch, not adapting one already existent. Does not model RCS and ACS but only main engines]

$$m_\mathrm{engines}(T) = m_\mathrm{b}\left[\lfloor x \rfloor + (x - \lfloor x \rfloor)^2\right]. \tag{30}$$

$$m_\mathrm{sys,dry}\left(m_\mathrm{prop}, T\right) = m_\mathrm{sys,prop}\left(m_\mathrm{prop}\right) + m_\mathrm{engines}(T). \tag{31}$$

B.15 Numerical values

Inputs (T in K, pressures in bar):

$R_m = 1.65$	$k = 3$ (MON-3)	$\varepsilon = 0$
$T_\mathrm{oxi} = T_\mathrm{fuel} = 273.15$	$T_\mathrm{pres} = 313.15$	hot case 313.15
$P_{u,i} = 20$	$\Delta P_{u,i} = 2$	$\Delta P_\mathrm{reg} = 5$
$P_{0,\mathrm{pres}} = 200$	$M_\mathrm{pres} = 0.004$ kg/mol	$T_\mathrm{b} = 1000$ N, $m_\mathrm{b} = 8.41$ kg

Results:

$\rho_\mathrm{oxi} = 1390.10$ kg/m³	$\rho_\mathrm{fuel} = 856.14$ kg/m³	$\rho_\mathrm{eq,prop} = 1125.27$ kg/m³
$a_\mathrm{tank,prop} = 0.88868$ L/kg	$1/\rho_\mathrm{eq,pres}$: 0.12956 L/kg	
$P_\mathrm{tank,i} = 22$ bar	$P_{f,\mathrm{pres}} = 27$ bar	$f_\mathrm{res} = 1.1561$
$M_\mathrm{eq,prop} = 1.0048$ kg/mol	$R_p = 3.9809 \times 10^{-3}$	$E_\mathrm{eq,pres} = 2591.2$ J/kg

Coefficients (with m_prop in kg, mass in kg):

	k_3	k_2	k_1	k_0
tank	1.9010×10^{-8}	-4.8730×10^{-5}	5.8910×10^{-2}	1.3192
bottle	–	4.8787×10^{-6}	6.5089×10^{-3}	2.9826
gas	–	–	3.9809×10^{-3}	–
total K	1.9010×10^{-8}	-4.3851×10^{-5}	6.9400×10^{-2}	4.3018

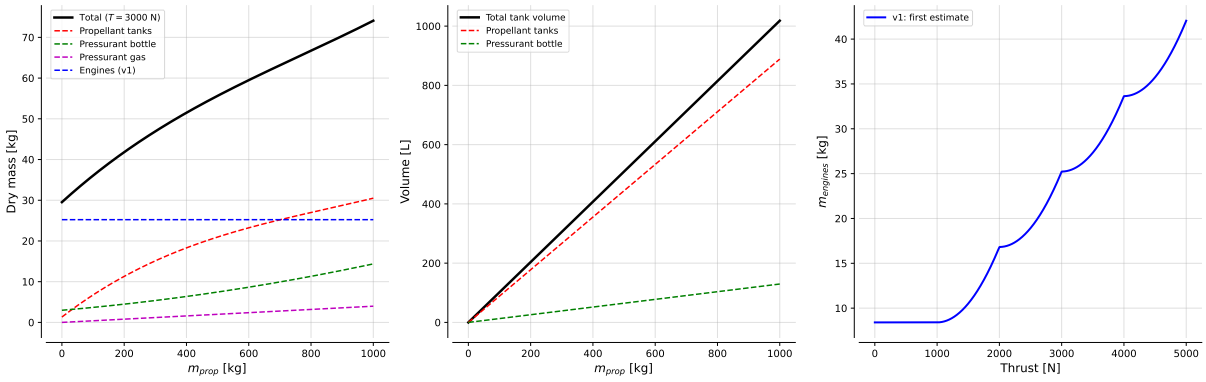


Figure 11: System dry mass as function of propellant mass and thrust, 2D slices

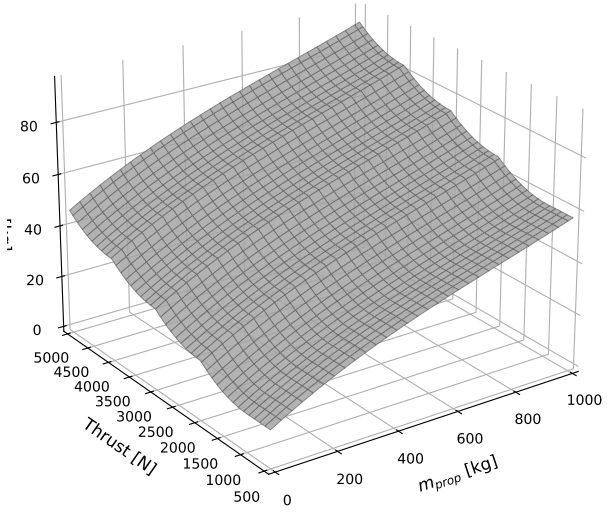


Figure 12: System dry mass as function of propellant mass and thrust, 3D plot

Appendix C

Avionics & GNC Requirements

Table 14: Avionics & GNC requirements (complete list). Levels: L4 = subsystem, L5 = component. Verification methods: A = analysis, I = inspection, T = test.

ID	Lvl	Requirement	Parent	Verif.	Status
GNC-01	L4	The GNC subsystem shall land the Hopper within a circle of 100 m diameter (50 m radius) centred on the designated target location.	L1-03	A	TBC
GNC-02	L4	The GNC subsystem shall navigate from liftoff to touchdown without ground support.	L1-03, L1-01	A	OK
GNC-03	L4	The GNC subsystem shall determine the Hopper initial position to ≤ TBD m and attitude to ≤ TBD deg (3σ) at the launch site before liftoff.	GNC-01, GNC-02	A	TBD
GNC-04	L4	The GNC subsystem shall estimate the Hopper attitude in all flight phases with an accuracy ≤ TBD arcsec (3σ).	GNC-01	A	TBD
GNC-05	L4	The GNC subsystem shall estimate position relative to the surface to ≤ TBD m (3σ) before the start of terminal descent, using on-board reference maps (TRN).	GNC-01	A	TBD
GNC-06	L4	The GNC subsystem shall estimate altitude above ground to ≤ TBD m and velocity relative to the surface to ≤ TBD m/s (3σ) from ≤ TBD km altitude down to touchdown.	GNC-01, GNC-08	A	TBD
GNC-07	L4	The GNC subsystem shall execute the ascent, coast and powered-descent profile defined by mission analysis, with navigation and steering losses ≤ TBD m/s.	L1-01, L1-02	A	TBD
GNC-08	L4	The GNC subsystem shall achieve touchdown with vertical velocity ≤ TBD m/s, horizontal velocity ≤ TBD m/s, tilt ≤ TBD deg and angular rates ≤ TBD deg/s.	L1-01 (safe delivery)	A	TBD

ID	Lvl	Requirement	Parent	Verif.	Status
GNC-09	L4	The GNC subsystem shall provide 3-axis attitude control in all flight phases with a pointing accuracy during burns ≤ TBD deg.	GNC-07	A	TBD
GNC-10	L4	The GNC subsystem shall measure altitude with at least two dissimilar sensors (e.g. laser altimeter + radar altimeter or Doppler LIDAR range channel).	GNC-06, GNC-08	A, T	OK
GNC-11	L4	The TRN reference maps shall be valid over the full descent altitude range (TBD km to TBD m), with resolution and illumination matching the conditions expected at the time of landing, and shall be stored on board (≥ TBD GB).	GNC-05	A	TBD
GNC-12	L4	The GNC subsystem should be able to use lunar navigation services (LunaNet / ESA Moonlight) when available.	GNC-01, GNC-03	R	OK
GNC-13	L4	The GNC subsystem shall perform ≥ TBD mid-course correction manoeuvre(s) during the ballistic arc, with a total correction $\Delta v \leq$ TBD m/s.	GNC-01, L1-02	A	TBD
GNC-14	L4	The landing shall take place with a Sun elevation at the target between TBD and TBD deg; departure shall take place within an illumination window compatible with the star trackers and TRN.	GNC-05, GNC-CAM-01, GNC-STR-01	A	TBD
AVI-01	L4	The avionics shall execute all GNC functions on board in real time, with a control loop rate ≥ TBD Hz.	GNC-02	A, T	TBD
AVI-02	L4	The avionics shall autonomously detect, isolate and recover from faults during the hop (FDIR), so that no single failure causes loss of the payload; safe-abort mode TBD.	GNC-02, L1-01	A, T	OK
AVI-03	L4	The avionics shall provide the payload data interface defined in the Payload ICD (RS-422 and Ethernet) at every payload position.	L1-09	I, T	TBC
AVI-04	L4	The avionics shall transfer payload data to the communication subsystem at a sustained rate ≥ the downlink rate of COM-01.	L1-10	T	OK
AVI-05	L4	The avionics shall store ≥ 100 GB (TBC) of payload and housekeeping data.	L1-10, L1-07	A	TBC

ID	Lvl	Requirement	Parent	Verif.	Status
AVI-06	L4	The avionics shall record the complete flight telemetry of each hop and retain it until downlinked.	GNC-02	T	OK
AVI-07	L4	The avionics shall operate for ≥ 72 h at the destination during lunar day. Lunar night survival is not required.	L1-07	A	OK
AVI-08	L4	The avionics shall operate within a temperature range of TBD to TBD °C and survive TBD to TBD °C (non-operating).	L1-07	T	TBD
AVI-09	L4	The avionics shall tolerate a total ionizing dose ≥ TBD krad and single-event effects for the mission duration without loss of function.	L1-01, L1-07	A	TBD
AVI-10	L4	The GNC & avionics mass shall be ≤ TBD kg and the power consumption ≤ TBD W per operating mode, including margin.	L1-04, L1-06	A, I	TBD
AVI-11	L4	The avionics shall use components with TRL ≥ 5 at PDR, preferring flight-proven equipment.	L1-05	R	OK
AVI-12	L4	The avionics shall accept and execute ground commands and provide housekeeping telemetry whenever a link is available.	L1-10	T	OK
AVI-13	L4	The avionics shall withstand the launch and landing mechanical environment (random vibration, sustained acceleration, acoustics) defined in the launcher payload user's guide.	L1-01	T	TBC
AVI-14	L4	Every GNC sensor and actuator shall be verified end-to-end in flight configuration before launch, including interlocks, safety switches and enable plugs, under representative conditions (lighting, regolith albedo, lidar back-scatter and multiple returns).	GNC-01	T, I	OK
AVI-15	L4	The avionics shall provide ≥ TBD independent release commands (power + signal) for payload hold-down and release mechanisms (HDRM).	L1-09	T	TBD
AVI-16	L4	The avionics shall acquire payload data at ≥ TBD Mbps from each high-rate payload position and write it to mass storage.	L1-09, AVI-05	T	TBD
AVI-17	L4	The avionics shall define a single-point (star) grounding scheme with a defined connection of signal ground, power return and chassis.	L1-01	R, T	TBC

ID	Lvl	Requirement	Parent	Verif.	Status
AVI-18	L4	All external electrical interfaces of the avionics (sensors, payload, umbilical, harness to propulsion) shall be protected against electrostatic discharge (ESD) up to TBD kV.	L1-01, L1-09	T	TBD
AVI-19	L4	The avionics shall command and monitor the switched, current-protected power lines of the power distribution unit (on/off, current, voltage telemetry) for every powered unit.	AVI-10, L1-06	T	TBD
AVI-20	L4	The avionics shall implement the on-board data network: RS-422 / UART for GNC sensors, CAN for housekeeping, high-rate links (Ethernet switch or SpaceWire router, TBD) between OBC units, cameras, payload and radio.	AVI-01, AVI-03, AVI-16	A, T	TBC
AVI-21	L4	The avionics shall acquire ≥ TBD temperature sensors and drive ≥ TBD heater / cooler channels for thermal control.	L1-07	T	TBD
AVI-22	L4	The avionics shall drive ≥ TBD propulsion valves and acquire ≥ TBD pressure and temperature transducers of the propulsion system.	L1-02, GNC-09	T	TBD
AVI-23	L4	The on-board software shall execute the hop autonomously on a preloaded trajectory, from commit-to-launch to touchdown; during the hop the ground can only command abort or safe mode.	GNC-02, AVI-02	A, T	TBC
AVI-24	L4	The flight software shall run on an existing, flight-proven real-time operating system (e.g. VxWorks), with custom applications (GNC, vision-based navigation, propulsion pressure control, FDIR) on top.	AVI-01, AVI-11	R	TBC
GNC-STR-01	L5	The star tracker shall provide attitude with an accuracy ≤ TBD arcsec (3σ) at ≥ TBD Hz, with Sun exclusion angle ≤ TBD deg.	GNC-04	A, T	TBC
GNC-IMU-01	L5	The IMU shall have gyro bias stability ≤ TBD deg/h and accelerometer bias stability ≤ TBD µg, such that the position error at start of descent is ≤ TBD m.	GNC-02, GNC-05	A	TBD
GNC-ALT-01	L5	The altimeters shall measure range to the surface from ≤ TBD km to touchdown with accuracy ≤ TBD m and shall be immune to interference from the other on-board active sensors.	GNC-06, GNC-10	A, T	TBD

ID	Lvl	Requirement	Parent	Verif.	Status
GNC-VEL-01	L5	The ground-relative velocity estimate (camera feature tracking fused with IMU and altimeters) shall provide 3-axis velocity relative to the surface from ≤ TBD km altitude with accuracy ≤ TBD cm/s.	GNC-06, GNC-08	A, T	TBD
GNC-CAM-01	L5	The TRN camera shall provide images with FOV TBD deg and resolution TBD m/px at TBD km altitude, under Sun elevations from TBD deg (pole) to TBD deg (equator).	GNC-05, GNC-11	A, T	TBD
AVI-OBC-01	L5	The on-board computer shall provide processing capacity ≥ TBD for state estimation, guidance, control, FDIR and data handling at the rate of AVI-01.	AVI-01	A, T	TBD
AVI-OBC-02	L5	The on-board computer shall be implemented as TBD (baseline: 3) redundant low-cost units with majority voting.	AVI-02, AVI-09	A, T	OK

Appendix D

Trajectory Optimization

Table 15: Boundary conditions, constraints and initial guesses of the single-phase and multiphase optimization.

Quantity	**Single phase**	**Multiphase**
Problem setup		
Objective	minimize initial mass $m(t_0)$, final mass fixed to m_{dry}	
Phases	1	5: vertical rise, ascent burn, coast, braking burn, terminal descent
Nodes	100	30 / 40 / 40 / 40 / 30
Transcription	one RK4 step per interval, piecewise constant controls, free final time	
Flight time	$t_f \in [60, 6000]$ s	$t_k \in [1, 6000]$ s per phase
Boundary conditions		
Initial state	$r = R$, $\theta = -89°$, $\dot{r} = 0$, $\dot{\theta} = 0$, m free	
Terminal state	$r = R$, $\theta = 0 \pm 10^{-4}°$, $\dot{r} = 0$, $\dot{\theta} = 0$, $m = m_{\text{dry}}$	$r = R$, $\dot{r} = 0$, $m = m_{\text{dry}}$
Events	–	end of vertical rise: $r = R + 50$ m start of terminal descent: $r = R + 50$ m, $\theta = 0$, $\dot{\theta} = 0$ phase linkage: $\boldsymbol{x}$ continuous
Path constraints		
Altitude	$0 \leq r - R \leq 400$ km	
Radial velocity	$\lvert\dot{r}\rvert \leq 2000$ m/s	
Angular rate	$\left\lvert\dot{\theta}\right\rvert \leq 2000 \text{ m/s } /R$	
Mass	$m_{\text{dry}} \leq m \leq m_{\text{dry}} + m_{\text{fuel}}$	
Thrust	$0 \leq T \leq T_{\max}$	vertical phases: $0 \leq T \leq T_{\max}$ burns: $0 < T \leq T_{\max}$ coast: $T = 0$
Thrust angle	$0 \leq \beta \leq \pi$	vertical phases: $\beta = \pi/2$ burns: $0 \leq \beta \leq \pi$
Initial guess		

Quantity	Single phase	Multiphase
Initial guess	analytic ballistic arc: θ linear, parabolic altitude (peak 250 km), $t_f = 800$ s, m linear from 900 kg to m_{dry}, $T = T_{\max}/2$, $\beta = \pi/2$	single-phase solution interpolated onto the phase grids; phase durations from preset breakpoints